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WHEEL HUB ASSEMBLY OR CRANK ASSEMBLY OR TRANSMISSION SYSTEM

Abstract

Disclosed is a wheel hub assembly comprising a hub housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with an externally arranged electric component of an electronic derailleur, the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the external electric component and one or more sensors and/or actuators.

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Background/Summary

FIELD

[0001] The invention relates to a transmission system for a vehicle, particularly a human powered vehicle, such as a bicycle, e.g. a two-wheeled bicycle.

BACKGROUND TO THE INVENTION

[0002] Transmission systems for bicycles are known. In bicycles, especially racing bicycles, the transmission system traditionally includes a front derailleur and a rear derailleur for shifting gears of the transmission system. An alternative to derailleurs is formed by gear hubs, where shifting of gears is accommodated by a gear shifting mechanism inside the, generally rear, wheel hub. A hybrid form is known where a gear hub torque transmission having at least two selectable gear ratios is coupled between the rear wheel hub and the rear sprocket. Herein the rear sprocket can include a plurality of gear wheels, selectable through a rear derailleur. Here the gear hub can take the place of a front derailleur.

[0003] Such gear hub gear shifting mechanisms can include one or more planetary gear sets. The planetary gear includes at least three rotational members, such as a sun gear, a planet carrier and a ring gear. A clutch system can be used for selectively coupling two of the rotational members, e.g. the planet carrier and the ring gear. When coupled, the hub gear shifting mechanism operates according to a first gear ratio. When decoupled, the hub gear shifting mechanism operates according to a second gear ratio.

[0004] Also gear hub shifting mechanisms are known wherein mechanisms are included in the gear hub for providing a plurality of different transmission ratios, such as five, seven or fourteen different gear ratios.

[0005] Many of these systems have in common that up- and downshifting is not always possible, depending on the riders pedal force. In some systems, it is required that the rider stops pedaling, or at least stops providing torque load to the system to allow up-shifting and/or down-shifting.

SUMMARY

[0006] It is an object to provide a transmission system, such as for a, e.g. two wheeled, bicycle. It is additionally or alternatively an object to provide an efficient transmission system, having minimal weight and minimal friction losses. Alternatively, or additionally, it is an object to enable, preferably electronically actuated, shifting of gears, wherein up- and downshifting should always be possible, not depending on the riders pedal force and/or electromotor torque.

[0007] According to an aspect is provided a wheel hub assembly for a bicycle, particularly for a distributed transmission system as described herein, comprising a hub housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with an externally arranged electric component, e.g. an external power source, external control unit and/or external user-operable shifter, the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the external electric component and one or more sensors and/or actuators. The one or more sensors and/or actuators may include, for example, a speed sensor, a position sensor, a brake sensor, a shift actuator, a torque sensor, a propulsion motor, and/or a brake actuator.

[0008] For example, provides a wheel hub assembly for a bicycle, comprising a hub housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with a, particularly rear, electronic derailleur,

the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the derailleur and the one or more sensors and/or actuators. Hence, the electric communication interface may be configured for operatively connecting the sensors and/or actuators held by the hub housing to, e.g. electronics of, a derailleur. Hence, sensors and/or actuators held by the hub housing may be communicatively and/or chargingly connected to the derailleur.

[0009] Optionally, the one or more sensor and/or actuators and the derailleur are powered by a common electric power source. The electric power source may for example include a generator and/or a battery. The electric power source may be held by the hub housing. Additionally or alternatively, the electric power source may be included by the derailleur, e.g. mounted to a derailleur body. In a particular example, the electric power source includes a battery that is either housed by the hub housing, or held by the derailleur.

[0010] Optionally, the electric power source includes a generator and/or a battery. The generator can be housed by the hub housing or can be arranged external to the housing, such as mounted to a bicycle frame or to the derailleur. The generator and the battery can be both housed by the hub housing. The generator can be housed by the hub housing, and the battery be arranged external to the hub housing, such as held by the derailleur. The generator and the battery can be both included by the rear derailleur, e.g. mounted to a derailleur body. The battery can be housed by the hub housing, and the generator can be arranged external to the hub housing, such as held by the derailleur or a bicycle frame. The generator may be configured to charge the battery. A power signal may hence be transmitted between the generator and the battery, such as via the electric connection interface.

[0011] Optionally, the derailleur includes the battery, e.g. mounted to a derailleur body, which battery can be connected to power the derailleur and the sensors and/or actuators held by the hub housing. A power signal may hence be transmitted from the battery at the derailleur, via the electric connection interface, to the one or more sensors and/or actuators held by the hub housing. Hence, no additional battery may be required within the hub housing.

[0012] Optionally, the hub housing may hold the battery for powering the one or more sensors and/or actuators, wherein the derailleur is powered by a battery held by the hub housing. A power signal may hence be transmitted from the battery held by the hub housing, via the electric connection interface, to, e.g. electronics of, the derailleur. Hence, no additional, e.g. dedicated, battery is needed at the derailleur for powering the derailleur.

[0013] Optionally, the hub assembly comprises an axle, such as a thru-axle, wherein a battery is mounted to or held by the axle for powering the one or more sensor and/or actuators held by the hub housing. The battery mounted to or held by the axle may additionally or alternatively be arranged for powering the derailleur.

[0014] Optionally, the hub housing includes a driver for being connected to a torque transfer member and a hub shell for being connected to a driven wheel, wherein the hub assembly comprises a hub transmission being selectively operable according to at least two different transmission ratios, and wherein the hub transmission is connected between the driver and the hub shell, and wherein the one or more sensors and/or actuators include a shift actuator for operating the hub transmission for switching from one transmission ratio of the at least two transmission ratios to another transmission ratio of the at least two transmission ratios. Accordingly, the aspect may particularly provide a hub assembly, e.g. for a distributed transmission system as described herein.

[0015] Hence, the hub assembly can be, for example wiredly, connected to a crank assembly of the distributed transmission system, which crank assembly includes, e.g. holds within the crank housing, the power source, e.g. battery, and/or a control unit for powering and/or controlling the hub shift actuator. The hub assembly may particularly include a transmission as described herein, e.g. one or more of the CVT, the first transmission, the second transmission and the third

transmission.

[0016] The hub assembly can for example be connected to a crank assembly arranged at a crank of the bicycle, which crank assembly includes, e.g. holds within the crank housing, a battery and/or a control unit for powering and/or controlling any sensor and/or actuator of the hub assembly via the electric connection interface.

[0017] Optionally, the electric connection interface is arranged for providing a wired electric connection between the one or more sensors and/or actuators and the external electric component. The external electric component may include one or more of an external control unit and/or an external power source such as a solar panel and/or battery, an external user-operable shifter, an external transmission such as a derailleur or crank transmission.

[0018] Hence, the aspect may particularly provide a wheel hub assembly for a bicycle, comprising a hub housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with a, particularly rear, electronic derailleur having a control unit associated therewith, the electric connection interface being arranged for transmitting electronic communication signals and/or electric power between the control unit and the one or more sensors and/or actuators.

[0019] Hence, the aspect optionally provides a wheel hub assembly for a bicycle, comprising a hub housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with a, particularly rear, derailleur having an electric power source, such as a battery and/or generator, associated therewith, the electric connection interface being arranged for transmitting electric power signals between the electric power source and the one or more sensors and/or actuators.

[0020] Optionally, the aspect particularly provides a wheel hub assembly for a bicycle, comprising a hub housing holding a control unit, and an electric connection interface for electrically connecting the control unit with a, particularly rear, derailleur, the electric connection interface being arranged for transmitting electronic communication signals between the control unit and the derailleur.

[0021] Optionally, the aspect particularly provides a wheel hub assembly for a bicycle, comprising a hub housing holding an electric power source, such as a battery and/or a generator, and an electric connection interface for electrically connecting the electric power source with a, particularly rear, derailleur, the electric connection interface being arranged for transmitting electric power signals between the electric power source and the derailleur.

[0022] Optionally, the connection interface comprises a first connection part having a first electric contact being electrically connected to the one or more sensors and/or actuators in the hub assembly; and a second connection part, being releasable connectable to the first connection part, and having a second electric contact arranged for being electrically connected to an external electric component, such as to an external power source and/or external control unit, wherein the first and second electric contacts are arranged for being in contact with one another so as to conduct electric signals therebetween when the first and second connection parts are mechanically coupled to each other. The first and second connection parts can e.g. form a plug/socket connection. The first and second connection parts can e.g. be maintained in mechanical and electric contact by magnetic attraction.

[0023] Optionally, the second connection part includes a battery electrically connected to the second contact. Hence, the battery held by the second connection part, may be electrically connectable to the sensors and/or actuators held by the housing, via the electric connection interface. Optionally, the second connection part includes a control unit. Hence, the control unit held by the second connection part, may be electrically connectable to the sensors and/or actuators held by the housing, via the electric connection interface.

[0024] Optionally, the hub assembly comprises a third connection part arranged for being mechanically, e.g. releasably, and electrically connected or connectable to the second connection

part.

[0025] Optionally, the third connection part includes an electrically conductive wire for providing a wired conduction path between the second electric contact and the external electric component. The electrically conductive wire may for example be releasably connectable to the second connection part by means of a plug/socket connection. Optionally, the electrically conductive wire is fixed to or integrated with the second connection part.

[0026] Optionally, the third connection part includes a wireless transmitter and/or receiver for providing a wireless communication channel between the third connection part and the external electric component. The wireless transmitter may for example be releasably connectable to the second connection part by means of a plug/socket connection. Optionally, the wireless transmitter is fixed to or integrated with the second connection part.

[0027] Optionally, the first and second electric contacts are arranged to provide at least two separate transmission channels between the first connection part and the second connection part, preferably at least three separate transmission channels, more preferably at least four separate transmission channels. Hence, multiple sensors and/or actuators held by the hub housing can separately be powered and/or be communicatively connected with external electrical components through the connection interface.

[0028] Optionally, the first connection part and the second connection part both include a through hole for accommodating a thru-axle therethrough. The thru-axle may be arranged for mounting the hub assembly, and a wheel associated therewith, to a frame of the bicycle.

[0029] Optionally, the first connection part is fixed to or integrated with a wheel axle, and the second connection part is fixed to or integrated with a hub axle end cap. The first contact and the second contact can be brought in contact with each other by attaching the end cap to the first connection part, e.g. the wheel axle, and can be released by removing the end cap therefrom.

[0030] Optionally, the first and second connection parts are coaxially arranged with respect to a wheel axis. The wheel axis may be defined as the axis about which the wheel associated with the hub assembly is rotatable.

[0031] Optionally, the first and second contacts are radially oriented with respect to the wheel axis. Hence, electric current can be conducted between the first and second contacts in radial direction with respect to the wheel axis. Optionally, the first and second contacts are axially oriented with respect to the wheel axis. Hence, electric current can be conducted between the first and second contacts in parallel to the wheel axis.

[0032] Optionally, the first contact is ring-shaped and extends around the wheel axis. Optionally, the second contact is ring-shaped and extends, in use when the first and second connection parts are connected, around the wheel axis. Hence, a contact face between the first and second contacts may accordingly be ring-shaped, and extending, in use, around the wheel axis.

[0033] Optionally, the first connection part and the second connection part are rotatable relative to one another about the wheel axis while the first contact and second contact are in contact.

[0034] Optionally, the first connection part includes an electrically conducting wire or wires extending from the hub housing, e.g. at or through an axle thereof having a first releasable connector at its free end, and the second connection part includes a second releasable connector for connection to the first releasable connector.

[0035] Optionally, the hub assembly comprises a bearing arranged between the driver and the wheel axle for allowing the driver to rotate relative to the wheel axle about the wheel axis, wherein the first connection part is arranged, in radial direction, between the bearing and the wheel axle. Hence, an electric conduction path extending between the first contact and the one or more sensors and/or actuators can be provided, running between the bearing and the wheel axle.

[0036] Optionally, the bearing is axially engaged by the second connection part.

[0037] Optionally, in use when the first connection part and the second connection parts are connected, a seal is arranged between the first connection part and the second connection part. The

seal may for example include a resilient O-ring coaxially provided between the first connection part and the second connection part, and extending around the wheel axis.

[0038] Optionally, the second contact is configured for being electrically connected to, e.g. electronics of, a rear derailleur. Hence, sensors and/or actuators held by the hub housing may be communicatively and/or chargingly connected to the rear derailleur. The rear derailleur may for example include a battery, e.g. mounted to a derailleur body, which battery can be connected to power the sensors and/or actuators held by the hub housing. Hence, no additional battery may be required within the hub housing. Alternatively, the derailleur may be powered by a battery held by the hub housing. Also, control signals may be transmitted between the derailleur and sensors and/or actuators held by the hub housing.

[0039] Optionally, the second contact is configured for being, e.g. directly, electrically connected to, e.g. electronics of, the crank transmission system. Hence, sensors and/or actuators held by the hub housing may be communicatively and/or chargingly connected to one or more sensors and/or actuators of the crank assembly.

[0040] Optionally, the second contact is configured for being electrically connected to a user-operable shifter. The user-operable shifter may for example be provided at handlebar of the bicycle, for allowing a user to review and/or operate the one or more sensors and/or actuators of the hub assembly.

[0041] According to an aspect is provided a crank assembly for a bicycle, particularly for a distributed transmission system as described herein, comprising a crank housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with an externally arranged electric component, e.g. an external power source, external control unit and/or external user-operable shifter, the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the external electric component and one or more sensors and/or actuators.

[0042] For example, the aspect provides a crank assembly for a bicycle, comprising a crank housing holding one or more sensors and/or actuators, and an electric connection interface for electrically connecting the one or more sensors and/or actuators with a, particularly rear, electronic derailleur, the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the derailleur and the one or more sensors and/or actuators. Hence, the electric communication interface may be configured for operatively connecting the sensors and/or actuators held by the crank housing to, e.g. electronics of, a derailleur. Hence, sensors and/or actuators held by the crank housing may be communicatively and/or chargingly connected to the derailleur.

[0043] The one or more sensors and/or actuators may include, for example, a speed sensor, a position sensor, a brake sensor, a shift actuator, a torque sensor, a propulsion motor, and/or a brake actuator.

[0044] Optionally, the crank assembly comprises an input shaft connected to a crank and an output shaft connected to the torque transfer member, and a crank transmission, being selectively operable according to at least two different transmission ratios, and wherein the crank transmission is connected between the input shaft and the output shaft, and wherein the one or more sensors and/or actuators include a shift actuator for operating the crank transmission for switching from one transmission ratio of the at least two transmission ratios to another transmission ratio of the at least two transmission ratios. Accordingly, the aspect may particularly provide a crank transmission system, e.g. for a distributed transmission system as described herein.

[0045] Hence, the crank assembly can be, for example be wiredly, connected to the hub assembly of the distributed transmission system, which hub assembly includes, e.g. holds within the hub housing, the power source, e.g. battery, and/or the control unit for powering and/or controlling any sensor and/or actuator of the crank assembly. It will be appreciated that, in addition or alternatively

to the crank shift actuator, the crank assembly may include other sensors and/or actuators, e.g. contained by the crank housing. The crank assembly may particularly include a transmission as described herein, e.g. one or more of the CVT, the first transmission, the second transmission and the third transmission.

[0046] Optionally, the input shaft and the output shaft are rotatable around a common drive axis, and the crank assembly comprises an electric propulsion motor for propelling the bicycle connected to the input shaft or the output shaft, wherein the electric propulsion motor has a rotatable output member that is rotatable about an electric motor output axis which extends transverse to the drive axis. Hence, a particular compact setup can be obtained.

[0047] Optionally, the electric motor is angularly spaced from the intermediate shaft.

[0048] According to an aspect is provided a transmission system for a vehicle, particularly a human powered vehicle such as bicycle. The transmission system comprises an electronically actuatable hub transmission having a first electronic actuator for selecting one of at least two different selectable transmission ratios, an electronically actuatable derailleur having a second electronic actuator for selecting one of a plurality of sprockets, and a common electric power source for providing electric power to both the first electronic actuator and second electronic actuator.

[0049] All options of the wheel hub assembly as recited above apply to the bicycle transmission. In particular, the following applies.

[0050] Optionally, the common electric power source includes a generator and/or a battery.

Optionally, derailleur includes the battery, e.g. mounted to a derailleur body. Optionally, a hub housing, housing the hub transmission and the first electronic actuator, holds the battery.

Optionally, the bicycle transmission comprises an axle, such as a thru-axle, wherein the battery is mounted to or held by the axle for powering the first and second electronic actuators. Optionally, the derailleur includes the generator. Alternatively, the hub housing holds the generator. It will be clear that when the common electric power source includes a common generator, multiple batteries may be used being charged by the common generator, such as a battery in the hub housing and a battery at the derailleur. Preferably, a single battery is used for providing electric power to both the first electronic actuator and second electronic actuator.

[0051] Optionally, the derailleur includes a control unit, e.g. mounted to a derailleur body, configured for controlling the first and second actuators. Optionally, the hub housing houses a control unit configured for controlling the first and second actuators.

[0052] According to an aspect is provided a transmission system for a vehicle, particularly a human powered vehicle such as bicycle, comprising a system input and a system output, wherein the system input is arranged to be connected to a power source, such as a crank and/or an electric motor and/or a user input, and the system output is arranged to be connected to a load, such as a driven wheel. The transmission system comprises a continuously variable transmission (CVT) and a first transmission between the system input and the system output, wherein the CVT and the first transmission are connected in series. The CVT is selectively operable according to a plurality of different transmission ratios within a continuous range of transmission ratios, and has an actuation element for switching the continuous variable transmission from one transmission ratio in said continuous range to another. The first transmission is selectively operable according to a first transmission ratio or a second transmission ratio. The first transmission can have a first clutch for switching the first transmission from the first transmission ratio to the second transmission ratio and/or vice versa. Optionally, the transmission system further includes a second transmission selectively operable according to a third transmission ratio or a fourth transmission ratio, and having a second clutch for switching the second transmission from the third transmission ratio to the fourth transmission ratio and/or vice versa.

[0053] The transmission system can be a crank transmission, included in a crank housing.

[0054] The transmission system can provide for a plurality of different system transmission ratios between the system input and the system output. The term “system transmission ratio” herein is

used to indicate the effective transmission ratio between the system input and the system output of the transmission system. The CVT particularly provides for a plurality of different continuously variable system transmission ratios between the system input and the system output.

[0055] The first transmission may have a first input and a first output, wherein the first input can be connected to the system input or wherein the first output may be connected to the system output. The CVT may have CVT input and a CVT output wherein the CVT input may be connected to the system input or wherein the CVT output may be connected to the system output. A particular arrangement provides that the first input is connected to the system input, that the CVT output is connected to the system output, and that the CVT input is connected to the first output.

Alternatively, the CVT input is connected to the system input, that the first output is connected to the system output, and that first input is connected to the CVT output.

[0056] Optionally, the first clutch is a form-closed clutch arranged to transfer torque in at least one rotational direction.

[0057] Optionally, the first clutch is a load-shifting clutch arranged for being coupled and/or decoupled under load. It may be preferred to couple and/or decouple the first clutch of the transmission system under load, to switch between transmission ratios while transmitting torque through the transmission system.

[0058] Optionally, the first transmission is arranged to operate according to the first transmission ratio when the first clutch is in a first state, and to operate according to the second transmission ratio when the first clutch is in a second state. The first clutch for instance has a coupled state in which a first clutch input and a first clutch output of the first clutch are coupled for transferring torque from the first clutch input to the first clutch output. The first clutch may also have a decoupled state in which the first clutch input and the first clutch output are decoupled. The first state of the first clutch may correspond to the coupled state, and the second state of the first clutch may correspond to the decoupled state, or vice versa.

[0059] Optionally, the first transmission includes a first transmission path and a second transmission path parallel to the first transmission path, wherein at least one of the first and second transmission paths includes the first clutch. Hence, torque can be selectively transmitted through either the first or the second transmission path, using the first clutch. For example, in a coupled state of the first clutch, torque can be transmitted through the transmission path which include the first clutch, e.g. the first transmission path. In an uncoupled state of the first clutch, no torque can be transmitted through the transmission path which includes the first clutch. Instead, torque can for instance be transmitted through the other, parallel, transmission path, e.g. the second transmission path.

[0060] The first transmission path may for example include a first gearing for providing the first transmission ratio; the second transmission path a second gearing for providing the second transmission ratio. Each of said gearings may for example include meshing gears, e.g. a meshing primary and secondary gear pair, and/or belt-driven wheels, e.g. a primary and secondary gear pair drivingly coupled via an endless drive member such as a belt or chain.

[0061] Optionally, the first clutch is arranged in the second transmission path, wherein the first transmission path includes a first freewheel clutch. The first freewheel clutch is optionally serially connected between the first transmission input and the first gearing. The first freewheel clutch may alternatively be serially connected between the first gearing and the first transmission output.

[0062] Optionally, the second transmission path includes a second freewheel clutch.

[0063] Optionally, an output of the second freewheel is connected to an input of the first clutch. Hence, freewheeling between the transmission input and transmission output can be allowed also when the first clutch is in the coupled state.

[0064] Optionally, the second freewheel clutch is serially connected between the second gearing and the first clutch.

[0065] Optionally, the first clutch is serially connected between the second freewheel clutch and

the second gearing.

[0066] Optionally, the second gearing is serially connected between the first clutch and the second freewheel clutch.

[0067] Optionally, the first transmission includes a planetary gear set.

[0068] Optionally the planetary gear set comprises a sun gear, a planet carrier carrying one or more planet gears, and a ring gear, wherein the planet carrier is coupled or couplable to an output of the CVT.

[0069] Optionally the sun gear is coupled or couplable to an output of the transmission system.

[0070] Optionally the one or more planet gears are each embodied as a stepped planet gear, having a large-radius planet gear member and a small-radius planet gear member corotatably connected to each other.

[0071] Optionally a planet gear rotation axis of the one or more planet gears extends parallel to a sun gear rotation axis of the sun gear, or a planet gear rotation axis of the one or more planet gears extends non-parallel, e.g. transverse, to a sun gear rotation axis of the sun gear.

[0072] Optionally the planetary gear set is arranged in the second transmission path for providing the second transmission ratio.

[0073] Optionally the first clutch is arranged in the second transmission path and the first transmission path includes a first freewheel clutch.

[0074] Optionally the second transmission path includes a second freewheel clutch.

[0075] Optionally, at least one of the second transmission ratio is a 1:1 transmission ratio.

Optionally, the smallest transmission ratio of the CVT and the first transmission is a 1:1 transmission ratio. Optionally, the smallest system transmission ratio of the transmission system is a 1:1 transmission ratio. For street or racing bicycles a 1:1 system transmission ratio as smallest system transmission ratio can be desirable. For mountain bikes or all terrain bikes a smallest system transmission ratio of smaller than 1:1 may be desirable.

[0076] Optionally, the CVT comprises a first drive element that is rotatable about a first CVT axis, a second drive element that is rotatable about a second axis parallel to the first axis, wherein the second drive element is movable relative to the first drive element in a direction transverse to the first axis; and first coupling elements for transferring torque between the first drive element and the second drive element, wherein the first coupling elements are provided at a constant first radius from the first axis and at a variable second radius from the second axis. Such CVT is e.g. described in co-pending application NL2028686 incorporated herein by reference in its entirety.

[0077] Optionally, the CVT comprises a third drive element that is rotatable about a third axis parallel to the second axis wherein the third drive element and the second drive element are movable relative to each other in a direction transverse to the third axis; and second coupling elements provided at a constant third radius from the third axis and at a variable fourth radius from the second axis, for transferring torque between the third drive element and the second drive element.

[0078] Optionally, the third drive element is directly couplable to the system output, for example by means of a one-way clutch.

[0079] Optionally, the system comprises a second transmission between the system input and the system output, wherein the continuously variable transmission, the first transmission and the second transmission are connected in series, the second transmission being selectively operable according to a third transmission ratio or a fourth transmission ratio, and e.g. having a second clutch for switching the second transmission from the third transmission ratio to the fourth transmission ratio and/or vice versa.

[0080] Optionally, the second transmission is arranged to operate according to the third transmission ratio when the second clutch is in a first state, and to operate according to the fourth transmission ratio when the second clutch is in a second state. The second clutch for instance has a coupled state in which a second clutch input and a second clutch output of the second clutch are

coupled for transferring torque from the second clutch input to the second clutch output. The second clutch may also have a decoupled state in which the second clutch input and the second clutch output are decoupled. The first state of the second clutch may correspond to the coupled state, and the second state of the second clutch may correspond to the decoupled state, or vice versa.

[0081] The first clutch and the second clutch are optionally, at least substantially, identical.

[0082] Optionally, the second transmission includes a third transmission path and a fourth transmission path parallel to the third transmission path, at least one of the third and fourth transmission paths including the second clutch. For example, in a coupled state of the second clutch, torque can be transmitted through the transmission path which include the second clutch, e.g. the third transmission path. In an uncoupled state of the second clutch, no torque can be transmitted through the transmission path which includes the second clutch. Instead, torque can for instance be transmitted through the other, parallel, transmission path, e.g. the fourth transmission path.

[0083] Optionally, the second clutch is arranged in the fourth transmission path, wherein the third transmission path includes a third freewheel clutch. The third freewheel clutch is optionally serially connected between the second transmission input and the third gearing. The third freewheel clutch may alternatively be serially connected between the third gearing and the second transmission output.

[0084] Optionally, the fourth transmission path includes a fourth freewheel clutch.

[0085] Optionally, an output of the fourth freewheel is connected to an input of the second clutch.

[0086] Hence, the second freewheel clutch may be serially connected to the first clutch at an input side of the first clutch and/or the fourth freewheel clutch may be serially connected to the second clutch at an input side of the second clutch.

[0087] Optionally, the fourth freewheel clutch is serially connected between the fourth gearing and the second clutch.

[0088] Optionally, the second clutch is serially connected between the fourth freewheel clutch and the fourth gearing.

[0089] Optionally, the fourth gearing is serially connected between the second clutch and the fourth freewheel clutch.

[0090] Optionally, in a particular compact setup, the second freewheel clutch is serially connected between the second gearing and the first clutch, and the fourth gearing is serially connected between the second clutch and the fourth freewheel clutch. Hence, an output of the first transmission may be formed by an output of the first clutch, and an input of the second transmission may be formed by an input the second clutch. The first clutch output and the second clutch input may for example be coupled to each other or integrated.

[0091] Optionally, the second transmission includes a third transmission path for providing the third transmission ratio and a fourth transmission path, parallel to the third transmission path, for providing the fourth transmission ratio, at least one of the third and fourth transmission paths including the second clutch.

[0092] Optionally, the third transmission ratio equals the lowest desired system transmission ratio by the first transmission ratio. For example, a product of the first transmission ratio and the third transmission ratio provides the lowest system transmission ratio.

[0093] Optionally, when a ratio of the second transmission ratio and the first transmission ratio is equal to U , a ratio of the fourth transmission ratio and the third transmission ratio is, e.g. within 5%, equal to $U^{sup.2}$. In other words, the ratio of the fourth transmission ratio and the third transmission ratio is, e.g. within 5%, equal to the square of the ratio of the second transmission ratio and the first transmission ratio. For example, when the second transmission ratio divided by the first transmission ratio is equal to U , a the fourth transmission ratio divided by the third transmission ratio is, e.g. within 5%, equal to $U^{sup.2}$.

[0094] Optionally, when a ratio of the second transmission ratio and the first transmission ratio is, e.g. within 5%, equal to U , a product of the first transmission ratio and the fourth transmission ratio is, e.g. within 5%, equal to $U_{sup.2}$. In other words, the product of the first transmission ratio and the fourth transmission ratio is equal to the ratio of the second transmission ratio and the first transmission ratio. For example, when the second transmission ratio divided by the first transmission ratio is, e.g. within 5%, equal to U , the first transmission ratio times the fourth transmission ratio is, e.g. within 5%, equal to $U_{sup.2}$.

[0095] Optionally, a ratio of the second transmission ratio and the first transmission ratio equals, e.g. within 5%, to a product of the second transmission ratio and the third transmission ratio.

[0096] Optionally, a ratio of the second transmission ratio and the first transmission ratio is between 1.1 and 1.3, preferably about 1.2. The ratio of the second transmission ratio and the first transmission ratio is for example 1.20 or 1.24. The ratio of the fourth transmission ratio and the third transmission ratio is for example 1.44 or 1.54. For example, the first transmission ratio is 1, the second transmission ratio is 1.2, the third transmission ratio is 1, and the fourth transmission ratio is 1.44.

[0097] Optionally, the second or fourth transmission ratio is a speed up transmission ratio. It will be appreciated that a transmission ratio of a transmission is defined as an output speed of an output of the transmission divided by an input speed of an input of the transmission. A speed up transmission ratio thus corresponds to a transmission in which the output speed of the transmission is higher than an input speed of the transmission. The speed-up transmission ratio is thus larger than one. Optionally, the first, second, third or fourth transmission ratio is a reduction transmission ratio. A reduction transmission ratio thus corresponds to a transmission in which the output speed of the transmission is lower than an input speed of the transmission. A reduction transmission ratio is smaller than one.

[0098] Optionally, the second transmission ratio is larger than the first transmission ratio. The first clutch may particularly be in the transmission path of the first transmission having the largest transmission ratio.

[0099] Optionally, at least one of the first transmission ratio and the second transmission ratio is a speed-down transmission ratio, and at least one of the third transmission ratio and the fourth transmission ratio is a speed-up transmission ratio. Alternatively, at least one of the first transmission ratio and the second transmission ratio is a speed-up transmission ratio, and at least one of the third transmission ratio and the fourth transmission ratio is a speed-down transmission ratio.

[0100] Optionally, the system of any preceding claim, wherein the second transmission ratio is smaller than the first transmission ratio.

[0101] Optionally, when the first transmission ratio or the second transmission ratio is, e.g. within 5%, equal to U , the third transmission ratio or the fourth transmission ratio is, e.g. within 5%, equal to $U_{sup.-2}$. In other words, the third transmission ratio or the fourth transmission ratio is, e.g. within 5%, equal to the squared inverse of the first transmission ratio or the squared inverse of the second transmission ratio.

[0102] Optionally, when the first transmission ratio or the second transmission ratio is, e.g. within 5%, equal to U , the third transmission ratio or the fourth transmission ratio is, e.g. within 5%, equal to $U_{sup.-1/2}$. In other words, the third transmission ratio or the fourth transmission ratio is, e.g. within 5%, equal to the square root of the inverse of the first transmission ratio or the square root of the inversed second transmission ratio.

[0103] Optionally, the transmission system comprises further transmissions, such as a third transmission, connected in series with the first and second transmissions between the input and the output, the third transmission having a third clutch, and the third transmission being operable according to a fifth transmission ratio and a sixth transmission ratio. The third clutch is optionally a form-closed clutch arranged to transfer torque in at least one rotational direction. The third clutch

may be arranged for being coupled and/or decoupled under load. It will be appreciated that the third transmission may be similar to the first transmission and/or the second transmission as described herein. Hence, any features described herein in view of the first and/or second transmission apply equally to the third transmission.

[0104] Optionally, the third transmission is arranged to operate according to the fifth transmission ratio when the third clutch is in a first state, and to operate according to the sixth transmission ratio when the third clutch is in a second state. The third clutch for instance has a coupled state in which a third clutch input and a third clutch output of the third clutch are coupled for transferring torque from the third clutch input to the third clutch output. The third clutch may also have a decoupled state in which the third clutch input and the third clutch output are decoupled. The first state of the third clutch may correspond to the coupled state, and the second state of the third clutch may correspond to the decoupled state, or vice versa.

[0105] Optionally, when a ratio of the second transmission ratio and the first transmission ratio is equal to U , a ratio of the fourth transmission ratio and the third transmission ratio is e.g., e.g. within 5%, equal to U , and a ratio of the sixth transmission ratio and the fifth transmission ratio is, e.g. within 5%, equal to U .

[0106] Optionally, the transmission system comprises a bypass transmission path between the input and the output parallel to the first and/or the second transmission, said bypass transmission path including a bypass transmission clutch, such as a freewheel clutch. Hence, the bypass transmission path can provide an additional transmission ratio between the system input and the system output.

[0107] Optionally, the transmission system comprises a bypass transmission path between the input and the output parallel to the first and/or the second and/or the third transmission, said bypass transmission path including a bypass clutch, such as a freewheel clutch.

[0108] Optionally, a bypass clutch actuator is provided for selectively actuating the bypass clutch between a coupled state in which the bypass clutch couples a bypass clutch input with a bypass clutch output for transferring torque and a decoupled state in which the bypass clutch input and bypass clutch are decoupled.

[0109] Optionally, the transmission system comprises an intermediate shaft, wherein the first transmission is operable between the input and the intermediate shaft, and the second transmission is operable between the intermediate shaft and the output. The first output of the first transmission and the second input of the second transmission may be connected or connectable to the intermediate shaft.

[0110] Optionally, the transmission system comprises an input shaft associated with the input, and an output shaft associated with the output, wherein the input shaft is connectable to the output shaft via the intermediate shaft.

[0111] Optionally, the output shaft extends coaxially to the input shaft. Hence, the input and output shafts can be substantially aligned.

[0112] Optionally, the output shaft is offset from the input shaft.

[0113] Optionally, each of the clutches, e.g. the first, second and third clutch, are a form-closed clutches arranged to transfer torque in at least one rotational direction.

[0114] Optionally, each of the clutches is a load-shifting clutch arranged for being coupled and/or decoupled under load.

[0115] Optionally, each load-shifting clutch has a clutch input, and a clutch output, each clutch including: [0116] a first unit connectable to the clutch input or clutch output, including at least one first abutment surface; [0117] a second unit connectable to the clutch output or clutch input, respectively, including at least one second abutment surface arranged for selectively engaging the first abutment surface, the first and second abutment surfaces being adapted to each other so as to allow disengaging under load, preferably in two directions; [0118] a third unit including at least one retaining member, the third unit being arranged for selectively being in a first mode or a second mode relative to the second unit, wherein the at least one retaining member in the first mode locks

the at least one second abutment surface for rotationally coupling the second unit to the first unit, e.g. in two directions, and in the second mode releases the at least one second abutment surface for decoupling the second unit from the first unit. The transmission system including such load-shifting clutch (or clutches) can be manufactured in a small form-factor suitable for integration in a two-wheeled bicycle.

[0119] Optionally, each clutch includes an actuator for moving the third unit from a first position to a second position or from a second position to a first position relative to the second rotatable unit.

[0120] Optionally, the third unit includes at least one actuation member arranged for moving the third unit from a first position to a second position or from a second position to a first position relative to the second rotatable unit.

[0121] Optionally, the clutch includes a first rotatable unit connectable to the input; a second rotatable unit connectable to the output; a third rotatable unit arranged for co-rotating with the second rotatable unit, the third rotatable unit being arranged for selectively being in a first rotational position or a second rotational position relative to the second rotatable unit, wherein the system is arranged for selectively in the first rotational position rotationally coupling the second rotatable unit to the first rotatable unit, and in the second rotational position decoupling the second rotatable unit from the first rotatable unit; wherein the system is arranged for temporarily changing rotation speed of the third rotatable unit relative to the second rotatable unit for rotating from the first position to the second position, or from the second position to the first position.

[0122] Optionally, each clutch further comprises a fourth unit including a selector, the selector being arranged for selectively being in a gripping or non-gripping mode; the selector in the gripping mode being arranged for gripping the at least one actuation member for rotating the third rotatable unit from the first position to the second position or from the second position to the first position relative to the second rotatable unit; the selector in the non-gripping mode being arranged for not engaging the at least one actuation member.

[0123] Optionally, the first unit of the first load-shifting clutch, and the first unit of the second load-shifting clutch are coupled or integrated together.

[0124] Optionally, the second unit of the second load-shifting clutch is bearing supported by the first unit of the first load-shifting clutch.

[0125] The clutches described herein may for example be load-shifting clutches as described in WO2018/199757A2, WO2020/085911A2, or WO2021/080431A1 which are hereby incorporated by reference in their entirety.

[0126] Optionally, the transmission system comprises a control unit configured for controlling one or more actuators of the bicycle in general, and of the transmission system in particular. The control unit may be configured to receive sensor signals from one or more sensors of the bicycle in general, and of the transmission system in particular. The control unit may for example control the one or more actuators based on the received sensor signals. The sensor signals may for example include a shift signal from a user-operable shifter, wherein the control unit controls the transmission system so as to shift gears of the transmission based on the received shift signal.

[0127] Optionally, the control unit is configured to receive a first shift signal and a second shift signal, and configured to control the CVT, particularly the CVT actuator and/or the first transmission, particularly the first transmission clutch, and/or optionally the second transmission, particularly the second transmission clutch (and/or the third transmission, e.g. via the third transmission clutch) for selecting a transmission ratio of the transmission system, e.g. by selectively coupling or decoupling the first clutch and/or actuating the CVT in response to receiving the first and/or second shift signal. The controller allows for simplified operation of the transmission system. Also, by having the CVT and first transmissions connected in series, there is no risk of lock-up of the transmission system when actuating the CVT, and the first clutch (and optionally the second and the third clutch) independently and/or simultaneously.

[0128] Optionally, the transmission system comprises a first housing holding the CVT and the first

transmission, wherein the control unit is arranged in the first housing. Hence, the first housing can include the CVT, the first transmission and the control unit. Optionally, the transmission system further comprises a second housing, separate from the first housing, holding the second transmission and/or the third transmission. The first housing may be configured to be arranged at or near a crank of the bicycle, such that the CVT and the first transmission together operate as a crank transmission. The first housing may correspond to a crank housing. The second housing may be configured to be arranged at or near a hub of a drive wheel of the bicycle, e.g. a rear wheel hub, such that the second transmission and/or the third transmission operate as a hub transmission. The second housing may hence correspond to a hub housing. It will be appreciated that the inverse configuration is also envisioned, i.e. that the first housing is arranged at the wheel hub and the second housing at the crank. The control unit is preferably held by either the first housing or the second housing. The control unit and the CVT are preferably contained by the same housing, e.g. the first housing or crank housing. The CVT and the first transmission are preferably contained by the same housing, e.g. the first housing or crank housing.

[0129] Optionally, the transmission system comprises a battery connected to power the control unit and/or one or more sensors and/or actuators of the transmission system. The battery is for example included in the first housing or the second housing of the transmission system, or externally arranged from the first and second housings. The battery, particularly when externally arranged, may optionally additionally power a propulsion motor of the bicycle. The transmission system may comprise a wired connection between the battery and the control unit and/or between the battery and one or more of the one or more sensors and/or actuators of the bicycle. The battery may optionally be non-rechargeable. The battery may also be non-replaceable. The power consumption of the sensors and/or actuators of the transmission system may for example be so low that the recharging of the battery is not needed, while a battery life time of the battery is sufficient to substantially correspond to the lifetime of other e.g. mechanical, parts of the transmission system.

[0130] Optionally, the transmission system comprises an actuator battery connected for powering, e.g. only, one or more transmission actuators of the transmission system, and an additional, separate, control unit battery connected for powering, e.g. only, the control unit. Each actuator optionally includes a respective actuator battery. The actuator battery may additionally, or alternatively, power, e.g. only, one or more sensors of the transmission system. The transmission actuator(s) may be configured to require less power than the control unit, such that the transmission actuator(s) may hence be provided with a dedicated actuator battery which is relatively small compared to the control unit battery. The actuator battery may for example be charged by locally harvested power, e.g. by a solar cell mounted to the bicycle and/or a generator. The actuator battery may be included in a common housing with the CVT, e.g. the first housing. The control unit battery may be provided external to the first and second housing, and may particularly correspond to a propulsion battery of the (e-)bicycle, which propulsion battery also powers an electric propulsion motor of the bicycle. The control unit battery, e.g. corresponding to the propulsion battery, may be charged separately from the actuator battery, e.g. by a connection to the mains.

[0131] Optionally, the control unit is arranged to be wirelessly connected to a user-operable shifter. The shifter may generate and wirelessly transmit signals to the controller. The control unit may accordingly comprise an input antenna for receiving a wireless shift signal, and one or more outputs for respectively controlling one or more actuators. Alternatively, the control unit is arranged to be wiredly connected to a user-operable shifter.

[0132] Optionally, the control unit comprises an electric connection interface for electrically, e.g. wiredly, connecting the control unit to any sensor and/or actuator of the transmission system of the bicycle. The electric connection may transmit electronic communication signals and/or electric power signals. For example, the CVT and the first transmission may be configured as a crank transmission of a bicycle located at or near a crank of the bicycle, wherein a further, e.g. separate, transmission, e.g. the second and/or third transmission, may be provided as a hub transmission at or

near a driven wheel hub of the bicycle. The electric connection may be configured to, e.g. wiredly, connect the control unit, which is part of a first one of the crank transmission or the hub transmission, to a second one of the hub transmission or the crank transmission. Hence, the control unit may receive and process a shift signal from a user-operable shifter, and in turn transmit a control signal to the hub transmission actuator and/or the crank transmission actuator to control operation of the hub transmission and/or the crank transmission based on the received shift signal. The electric connection between control unit and the transmission actuator is preferably wired, and is optionally releasable. Alternatively, the control unit may comprises a wireless connection interface for wirelessly connecting the control unit to any transmission actuator of the bicycle.

[0133] Optionally, the first shift signal is an upshift signal and the second shift signal is a downshift signal, and the controller is configured to selectively control the CVT and/or the first (and/or second and/or third) clutch for selecting the next higher system transmission ratio in response to receiving the upshift signal, and for selecting the next lower system transmission ratio in response to receiving the downshift signal. Hence, the rider only needs to provide the upshift signal or the downshift signal, e.g. by means of one or more controls, levers, switches or the like. Preferably, the first and second shift signals are electronic signals. The first and/or second and/or third clutch can include a first and/or second and/or third actuator, respectively, for electrically actuating the respective clutch to couple or decouple. The controller then controls the first and second (and third) actuators in response to the upshift or downshift signal provided by the rider. Depending on the system transmission ratio used at that point in time, the next higher system transmission ratio can be achieved by actuating the first actuator and/or the second actuator (and/or third actuator). The controller is configured to select and actuate the appropriate actuator. Thus, shifting is simplified for the user.

[0134] Optionally, the first shift signal is an upshift signal and the second shift signal is a downshift signal, and the controller is configured to selectively control the CVT and the first clutch (and/or second clutch and/or third clutch) for selecting the second next, third next, fourth next higher or lower system transmission ratio in response to receiving a bail-out signal, the bail-out signal e.g. comprising the upshift signal and downshift signal at the same time, or within a specified time-interval, typically smaller than 1 s.

[0135] Optionally, the first and second shift signals are wireless signals, and wherein the control unit is arranged to receive the wireless shift signals.

[0136] Optionally, the system comprises one or more actuators, particularly one or more electric actuators, arranged for actuating the CVT and the first clutch (and second and third clutches).

[0137] Optionally, the one or more actuators are operatively connected to the control unit.

[0138] Optionally, the control unit and the one or more actuators are arranged to wirelessly communicate. Optionally, the control unit and the one or more actuators are arranged to wiredly communicate. It will be appreciated that combination of wireless and wired communication is envisioned.

[0139] Optionally, the system comprises a torque sensor for measuring an input torque at the input, wherein the torque sensor is operatively connected to the control unit. The torque sensor may for example be arranged to measure a torque at a crank and/or at a crankshaft.

[0140] Optionally, the torque sensor is integrated in the transmission system.

[0141] Optionally, the control unit and the torque sensor are arranged to wirelessly communicate.

[0142] Optionally, the torque sensor is arranged to be powered by a rotary motion of a crank and/or crankshaft about a crank axis.

[0143] Optionally, the torque sensor is arranged to be wirelessly powered.

[0144] Optionally, the system comprises an electric motor, for propelling, or assist propelling, of the vehicle. Optionally, the electric motor is connected to the CVT input, Optionally, the electric motor is connected between the CVT and the first transmission. Optionally, the electric motor is connected to the system output. Optionally, the electric motor is connected between the first

transmission and the second transmission.

[0145] Optionally, the electric motor is connected between the first transmission and the system output. Optionally, the electric motor is connected between the second transmission and the system output.

[0146] Optionally, the system comprises a battery arranged for powering the electric motor, and further arrange for powering the one or more actuators and/or sensors.

[0147] It will be appreciated that the CVT can be any type of CVT. Optionally, the continuously variable transmission is of a ratcheting type, e.g. using freewheel or one-way drive modules.

Optionally, the continuously variable transmission is of a friction type, toroidal type, or pulley type.

[0148] According to a further aspect is provided a distributed transmission system, comprising a hub assembly as described herein, and an electronic derailleur transmission. According to a further aspect is provided a distributed transmission system, comprising a crank assembly as described herein, and an electronic derailleur transmission. According to a further aspect is provided a distributed transmission system, comprising a hub assembly as described herein, a crank assembly as described herein, and an electronic derailleur transmission.

[0149] According to a further aspect is provided a distributed transmission system, comprising a crank assembly including the CVT and the first transmission as described herein, and a hub assembly comprising the second transmission and/or the third transmission as described herein. The CVT and the first transmission can be included in a common housing, such as a crank housing. The second transmission and/or the third transmission can be included in a separate hub housing, e.g. at least partly formed by a wheel hub shell, separate from the crank housing.

[0150] According to a further aspect is provided a distributed transmission system, comprising a crank assembly including the second transmission and/or the third transmission as described herein, and a hub assembly comprising the CVT and the first transmission as described herein. The CVT and the first transmission can be included in a common housing, e.g. at least partly formed by the wheel hub shell. The second transmission and/or the third transmission can be included in a common crank housing.

[0151] According to a further aspect is provided a distributed transmission system, comprising a crank assembly including the CVT as described herein, and a hub assembly comprising the first transmission and/or the second transmission and/or the third transmission as described herein. The CVT can be included in a crank housing. The first transmission and/or the second and/or third transmissions can be included in a common hub housing, e.g. at least partly formed by the wheel hub shell, separate from the crank housing.

[0152] According to a further aspect is provided a distributed transmission system, comprising a crank assembly including the first transmission and/or the second transmission as described herein, and a hub assembly comprising the CVT as described herein. The CVT and the first transmission can be included in a common housing, e.g. the wheel hub shell. The first and/or second transmissions can be included in a crank housing.

[0153] Optionally, the distributed transmission system includes a control unit, e.g. as described herein, configured for receiving a shift signal from a user-operable shifter device, and for controlling the hub transmission and/or the crank transmission based on the received shift signal.

[0154] Optionally, the crank housing includes the control unit, which control unit is configured for controlling the crank transmission, and is further, e.g. wiredly or wirelessly, connected to the hub assembly for controlling the hub transmission. Alternatively, the hub housing may include the control unit, which control unit is configured for controlling the hub transmission, and is further, e.g. wiredly or wirelessly, connected to the crank transmission system for controlling the crank transmission. It will also be appreciated that, alternatively, both the crank housing and the hub housing may include a respective control unit.

[0155] Optionally, the crank housing includes a battery, e.g. as described herein, which battery is connected to power a sensor and/or actuator of the crank transmission and which battery is further

connected to the hub assembly for powering a sensor and/or actuator of the hub transmission. The hub housing, hence, need not to contain a battery. It may be preferred to provide the battery, being a relatively heavy component, within the crank housing, to improve handling of the bicycle. The crank housing may include a charging port, for allowing the battery to be charged by an external source. The crank housing may also include an electric propulsion motor, e.g. being powered by the battery. Alternatively, the hub housing may include a battery as described herein, which battery is connected to power a sensor and/or actuator of the hub assembly and is further is connected to the crank transmission system for powering a sensor and/or actuator of the crank transmission system. The crank housing, hence, need not to contain a battery. It will also be appreciated that both the crank housing and the hub housing may include respective batteries as described herein. The battery of the crank transmission and/or the hub transmission may be rechargeable, e.g. by connection to the mains or locally by an energy harvesting device such as a solar panel. The battery may alternatively be non-rechargeable. The battery may alternatively be provided external to the crank housing and the hub housing.

[0156] Optionally, only one of the crank housing and the hub housing includes the battery, and only another one of the crank housing and the hub housing includes the control unit, wherein the battery is connected to power the control unit, e.g. through a wired connection.

[0157] Optionally, the distributed transmission system may comprise a wired connection between the crank assembly and the hub assembly, particularly between on the one hand the control unit and/or battery being held by one of the crank housing and the hub housing, and on the other hand the one or more sensors and/or actuators being held by another one of the crank housing and the hub housing. The electric connection may hence be wired, but may alternatively or additionally be wireless. For example, the control unit and the one or more sensors and/or actuators may be wirelessly connected, whereas the battery and the sensors and/or actuators are wiredly connected. One or both of the crank assembly and the hub assembly may for example comprise a connector interface for releasably connecting an electric wire so as to directly connect the hub assembly with the crank assembly. Hence a modular distributed transmission system may be obtained, in which multiple transmission can easily be communicatively connected to each other.

[0158] According to a further aspect is provided a bicycle comprising a transmission system, e.g. a distributed transmission system, as described herein, or a hub assembly and/or a crank assembly as described herein. The bicycle may for example comprise a distributed transmission system as described herein.

[0159] Optionally, the bicycle comprises a torque transfer system having a torque transfer member, such as a chain or belt or shaft, wherein a crank drives an input of the torque transfer system, and wherein an output of the torque transfer system drives a driven wheel of the bicycle, wherein the transmission system is arranged between the crank and the input of the torque transfer system. The CVT and the first transmission (and optionally the second transmission and the third transmission) may be housed in a common housing placed at the location of the crank.

[0160] Optionally, the bicycle comprises a torque transfer system having a torque transfer member, such as a chain or belt or shaft wherein a crank drives an input of the torque transfer system, and wherein an output of the torque transfer system drives a driven wheel of the bicycle, wherein the transmission system is arranged between the output of the torque transfer system and a wheel hub of the driven wheel. The CVT and the first transmission (and optionally the second transmission and the third transmission) may be housed in a common housing at the wheel axle.

[0161] Optionally, the bicycle comprises a torque transfer system having a torque transfer member, such as a chain or belt or shaft wherein a crank drives an input of the torque transfer system, and wherein an output of the torque transfer system drives a driven wheel of the bicycle, wherein the CVT and the first transmission of the transmission system are arranged between the crank and the input of the torque transfer system, and wherein the second transmission of the transmission system is arranged between the output of the torque transfer system and a wheel hub of the driven wheel.

The CVT and the first transmission may be housed at the location of the crank, and the second transmission may be housed at the wheel axle.

[0162] Optionally, the bicycle comprises a torque transfer system having a torque transfer member, such as a chain or belt or shaft wherein a crank drives an input of the torque transfer system, and wherein an output of the torque transfer system drives a driven wheel of the bicycle, wherein the CVT and the first transmission of the transmission system are arranged between the crank and the input of the torque transfer system, and wherein the second transmission of the transmission system is arranged between the output of the torque transfer system and a wheel hub of the driven wheel. The CVT and the first transmission may be housed in a common housing placed at the location of the crank, and the second transmission may be housed at the wheel axle.

[0163] Optionally, the CVT is arranged between the first transmission and the second transmission. Optionally, the CVT is arranged between the second transmission and the third transmission.

Optionally, the CVT is arranged between a system input and the first transmission or between the second transmission and a system output. Optionally, the CVT is arranged between a system input and the first transmission or between the third transmission and a system output. The CVT can be a ratcheting type of CVT, e.g. using freewheel or one-way drive modules. The CVT can be used for increasing the number of system transmission ratios. The CVT can be controlled to selectively operate at one of two or three (or more) distinct transmission ratios. The CVT can be controlled to operate at a first CVT transmission ratio and a second CVT transmission ratio. A ratio of the second CVT transmission ratio and the first CVT transmission ratio can be chosen to be, e.g.

approximately, half of the ratio of the second transmission ratio and the first transmission ratio. The CVT can be controlled to operate at a first CVT transmission ratio, a second CVT transmission ratio and a third CVT transmission ratio. A ratio of the second CVT transmission ratio and the first CVT transmission ratio can be chosen to be, e.g. approximately, one third of the ratio of the second transmission ratio and the first transmission ratio, and a ratio of the third CVT transmission ratio and the first CVT transmission ratio can be chosen to be, e.g. approximately, two third of the ratio of the second transmission ratio and the first transmission ratio.

[0164] The CVT may for example comprise a first drive element that is rotatable about a first axis; a second drive element that is rotatable about a second axis parallel to the first axis, wherein the first drive element and the second drive element are movable relative to each other in a direction transverse to the first and second axis; and first coupling elements provided at a constant first radius from the first axis and at a variable second radius from the second axis, for transferring torque between the first drive element and the second drive element.

[0165] The CVT may be configured to be operable according to any transmission ratio within a continuous range of CVT transmission ratios. The CVT may particularly be configured to be operable according to a predetermined finite set of transmission ratios within the continuous range of CVT transmission ratios.

[0166] A transmission ratio of a transmission as described herein, is particularly an output speed of the transmission divided by an input speed of the transmission.

[0167] According to an aspect is provided a gearless transmission unit such as for a bicycle, providing at least two discrete selectable transmission ratios, wherein a first of the at least two transmission ratios is provided by a first endless drive member, and wherein a second of the at least two transmission ratios is provided by a second endless drive member.

[0168] Optionally, the first and second endless drive members are placed in parallel between an input and an output of the gearless transmission unit, and the gearless transmission unit includes a selector for selecting power transmission via the first or the second endless drive member.

[0169] Optionally, the gearless transmission unit includes a clutch for selecting power transmission via the first endless drive member or the second endless drive member.

[0170] Optionally, the gearless transmission unit further includes a third endless drive member and a fourth endless drive member, wherein the third and fourth endless drive members are placed in

parallel between an output of the first and second endless drive members and an output of the gearless transmission unit, and the gearless transmission unit includes a selector for selecting power transmission via the third or the fourth endless drive member.

[0171] Optionally, the gearless transmission unit includes a clutch for selecting power transmission via the third endless drive member or the fourth endless drive member.

[0172] Optionally, the clutch is arranged to be coupled and/or decoupled under load. The clutch is for example a load-shifting clutch.

[0173] Optionally, at least one of the first, second, third and fourth endless drive members is non-lubricated. In particular, each endless drive member of the gearless transmission unit may be non-lubricated. Hence, no lubrication fluid is provided on at least one of the first endless drive member, the second endless drive member, the third endless drive member and the fourth endless drive member, particularly on all four of said endless drive members. A dry drive system can hence be obtained.

[0174] Optionally, the at least one of the first, second, third and fourth endless drive members comprises, e.g. is, a dry belt or a dry chain.

[0175] Optionally, at least one of the first, second, third and fourth endless drive members comprises a lubricated chain. In particular, and as alternative to a dry drive system, each endless drive member of the gearless transmission unit may be lubricated, e.g. with a lubrication fluid such as an oil.

[0176] Optionally, the gearless transmission unit includes a continuously variable transmission.

[0177] According to an aspect, each of the transmissions provided by the transmission system as described herein is oil-free, preferably lubrication-free.

[0178] According to an aspect is provided a hub assembly for a bicycle including the gearless transmission unit.

[0179] According to an aspect is provided a crank assembly for a bicycle including the gearless transmission unit.

[0180] According to an aspect is provided a bicycle including the gearless transmission unit.

[0181] It will be appreciated that any one or more of the above aspects, features and options can be combined. It will be appreciated that any one of the options described in view of one of the aspects can be applied equally to any of the other aspects. It will also be clear that all aspects, features and options described in view of the transmissions system apply equally to the bicycle. It will also be clear that all aspects, features and options described in view of the control device and control system apply equally to the transmission system, assembly, and bicycle. It will be appreciated that any aspects, options and feature described in relation to the hub assembly apply equally to the crank assembly.

Description

BRIEF DESCRIPTION OF THE DRAWING

[0182] The invention will further be elucidated on the basis of exemplary embodiments which are represented in a drawing. The exemplary embodiments are given by way of non-limitative illustration. It is noted that the figures are only schematic representations of embodiments of the invention that are given by way of non-limiting example.

[0183] In the drawing:

[0184] FIGS. 1A-1B show schematic examples of a transmission system;

[0185] FIGS. 2A-2B show a schematic example of a transmission system;

[0186] FIGS. 3A-3B show a schematic example of a transmission system;

[0187] FIGS. 4A-4B show schematic layouts of a transmission system;

[0188] FIGS. 5A-5B show schematic layouts of a transmission system;

[0189] FIGS. **6A-6B** show schematic layouts of a transmission system;
[0190] FIGS. **7A-7B** show a schematic example of a transmission system;
[0191] FIGS. **8A-8B** show schematic layouts of a transmission system;
[0192] FIGS. **9A-9B** show schematic layouts of a transmission system;
[0193] FIGS. **10A-10B** show schematic layouts of a transmission system;
[0194] FIGS. **11A-11C** show an example of a transmission system for a crank assembly;
[0195] FIG. **12** shows a schematic example of a transmission system;
[0196] FIG. **13** shows a schematic example of a transmission system;
[0197] FIG. **14** shows a schematic example of a transmission system;
[0198] FIGS. **15-19** show a cross sectional view of a hub assembly;
[0199] FIGS. **20A-20B** show schematic examples of a bicycle transmission system;
[0200] FIG. **21** shows a schematic example of a transmission system;
[0201] FIG. **22** shows a schematic example of a transmission system;
[0202] FIG. **23** shows an example of a bicycle; and
[0203] FIG. **24** shows an example of a bicycle.

DETAILED DESCRIPTION

[0204] FIG. **23** shows a bicycle **1000**. The bicycle **1000** comprises a frame **1002** with a front fork **1005** and a rear fork **1007**, as well as a front wheel and a rear wheel **1011**, **1013** located in the front and rear fork respectively. The bicycle **1000** further comprises a crank **1017**, and a front chain wheel **1019**. The bicycle **1000** also comprises a set of rear sprockets **1021** and a rear wheel hub **1022** of the rear wheel **1013**. A chain or belt **1023** engages over the front chain wheel **1019** and a selective one of the rear sprockets of the cassette **1021** to form an offset drive **908**. The bicycle **1000** comprises a bicycle transmission system **1**. Here, the transmission system **1** comprises a rear derailleur **480** for shifting the chain **1023** from one chain to another. In this example, a hub transmission is provided, interconnected between the cassette of rear sprockets **1021** and the rear wheel hub **1022**. The hub transmission is selectively operable according to a plurality of transmission ratios.

[0205] FIGS. **20a-2C** show schematic examples of a bicycle transmission system **1**, comprising a transmission **907** housed by a housing **490**. The transmission **907** may for example be embodied as a first transmission **100** or a second transmission **200** as described hereinbelow, or any other transmission. The transmission **907** may particularly comprise a planetary gear set. The transmission **907** in this example acts as a hub transmission, e.g. second transmission **200**, such as for the bicycle **1000** shown in FIG. **23**. The transmission **200** here provides multiple selectable transmission ratios between a driver **906** and a hub shell **902** of the bicycle rear wheel **1013**. The housing **490** may hence be formed, at least in part, by the hub shell **902** and/or the driver **906**. Alternatively, the transmission **907** may act as a crank transmission, e.g. first transmission **100**, e.g. providing multiple selectable transmission ratios between the crank **1017** and a front chain wheel **1019** of the bicycle **1000**.

[0206] FIGS. **15-19** show cross-sectional views of a hub assembly **900** for a bicycle **1000**. In this example, the hub assembly is a rear wheel hub assembly. The hub assembly **900** here includes a hollow axle **901** extending along a wheel axis **903**. In this example, the hollow axle **901** is arranged for non-rotatably being fixed to a frame, e.g. a bicycle frame. The hub assembly **900** includes a hub shell **902**. Here the hub shell **902** is provided with apertures **904**, e.g. for connection of spokes of a wheel. The hub assembly **900** further includes a driver **906**. The driver **906** in this example is arranged for receiving a cassette of gear wheels (not shown). The hub shell **902** and the driver **906** form, at least partly, a hub housing of the hub assembly **900**.

[0207] The hub assembly **900** can comprise various sensor and/or actuators, e.g. a speed sensor, torque sensor, brake actuator, shift actuator, etc. In this example, the hub assembly includes a transmission system, having a transmission **907** operable between the driver **906** and the hub shell **902** according to at least two different transmission ratios. The transmission **907** includes, here, a

planetary gear set **910**, The planetary gear set **910** includes a sun gear **912**, a planet carrier **914** with planet gears **916** and a ring gear **918**. In this example, a clutch **C1** is arranged so as to selectively couple two of the sun gear, the planet carrier and the ring gear. In this example, the clutch **C1** is arranged to selectively couple the planet carrier **914** and the ring gear **918**. The planet carrier **914** is also fixedly coupled to the hub shell **902**. Therefore, depending on whether the planet carrier **914** and ring gear **918** are rotationally coupled, or rotationally uncoupled, driving the driver **906** causes the hub shell **902** to rotate according to a first or second transmission ratio relative to the driver **906**. An overrunning clutch may be positioned between the sun gear **912** and the axle **901**.

[0208] The clutch **C1** can e.g. be used for selectively operating a planetary gear according to a first transmission ratio when the planet carrier **914** is engaged with the ring gear **918**, and according to a second transmission ratio when the planet carrier **914** is disengaged from the ring gear **918**. The clutch **C1** is here electrically operable. It will be appreciated that a plurality of transmission ratios may be obtained using the planetary gear set and an electrically operable clutch or brake in alternative ways, e.g. by selectively coupling or decoupling a sun gear with the, non-rotating, axle **901**.

[0209] The hub assembly comprises an electric shift actuator **504**. Here, an electric clutch actuator **920** for moving the clutch between a coupled state in which it couples the planet carrier **914** to the ring gear **918**, and a decoupled state in which it decouples the planet carrier **914** from the ring gear **918**. The electric clutch actuator **920**, and/or any additional or alternative sensor and/or actuator held by the hub housing, may be communicatively connected to an external control unit, here via a wire **926**. Also, the electric clutch actuator **920**, and/or any additional or alternative sensor and/or actuator held by the hub housing, may be electrically powered, e.g. by an external battery, here via the wire **926**, or via an additional dedicated charging wire. The wire **926** forms, at least partly, an conduction path between the clutch actuator **920** being held internal to the hub housing, and an external electric component, e.g, an external power source or control unit, being arranged external of the hub housing.

[0210] The assembly comprises an electric connection interface, including a first connection part **921** and a second connection part **922**. The first and second connecting parts **921**, **922** are releasably connectable to each other. Hence, the second connection part **922** may for example be physically separated from the first connection part **921**. The first connection part **921** comprises a first electric contact **931**, and the second connection part comprises a second electric contact **932**. When the first connection part **921** and the second connection part **922** are connected to each other, the first and second contacts **931**, **932** are in electrically conductive contact with each other for transmitting electric signals between the first connection part **921** and the second connection part **922**. The second contact **932** is electrically connected to the wire **926**. The first contact **931** is electrically connected to the clutch **C1**, or the actuator operating the clutch. Here, the first contact **931** and the second contact **932** are configured to provide four separate transmission channels between the first and second connection parts **921**, **922**, but it will be appreciated that any other number of transmission channels may also be possible. It will be appreciated that alternatively the wire **926** may be directly electrically connected to the actuator or other electronics inside the hub assembly.

[0211] FIGS. **16-19** shows close-up views of the first and second connection parts **921**, **922**. The first connection part **921** is here formed by a bush connected to the hollow wheel axle **901**. It will be appreciated that the first connection part **921** may also be integral with the axle **901**. The second connection part **922** is here formed by an end cap of the hub axle. A bearing **936**, here a ball bearing, is arranged between the driver **906** and the axle **901**. The first connection part **921** is particularly provided between the bearing **936** and the axle **901**. Hence, an electric signal path can be provided extending between the first electric contact **931** and the clutch **C1**, which signal path runs between the bearing **936** and the axle **901**. A seal **935** is provided between the first and second connection part **921**, **922**.

[0212] In FIG. 16, the wire 926 extends from the second connection part 922, here an end cap of the hub axle. The wire 926 is particularly fixed to the second connection part 922.

[0213] FIG. 17 shows an example wherein the second connection part 922 is provided with a socket connector 923 for releasably connecting to one or more complementary plug connectors 924. It will be appreciated that the second connection part 922 may conversely be provided with a plug connector. The plug connector 924 is wiredly connected, e.g. to an external control unit, and/or an external battery. In this example two wires 926 are releasably connectable to the second connection part 922.

[0214] FIG. 18 shows an example wherein a wireless transceiver 925 is connectable to the second connection part 922, via the plug/socket-connection 923,924. A wireless communication channel may hence be provided between the second connection part 922 and an external control unit.

[0215] FIG. 19 shows an example wherein the wireless transceiver 925 is fixed to or integrated with the second connection part 922.

[0216] FIGS. 20A-20C show schematic examples of the hub assembly 900, such as of FIGS. 15-19, being connected, e.g. via the electric connection interface, to an electronic rear derailleur 480. The rear derailleur 480 is associated with an offset drive 908, comprising a front chain wheel 1023 connected via a chain or belt 1019 to a cassette of rear sprockets 1021. The rear derailleur 480 is particularly configured to shift the chain or belt 1019 from one sprocket of the cassette 1021 to another, so as to obtain a different transmission ratio with the offset drive 908. The rear derailleur 480 is particularly electrically powered and electronically controlled. The transmission 907 is connected in series to the offset drive 908. In an example, the transmission 907 is a hub transmission operable according to at least two different transmission ratios. The transmission 907 includes or is associated with an electronic shift actuator 504. The shift actuator 504 is arranged for shifting the transmission 907 from one transmission ratio to another, for example by activating and/or deactivating one or more clutches of the transmission 907. The shift actuator 504 is held by the housing 490 in which also the transmission 907 is housed. In this example, the housing 490 can be formed by a hub housing and/or driver.

[0217] In the example of FIGS. 20A and 20B, the rear derailleur 480 and the shift actuator 504 are powered by a common power supply 507. In this example, the power supply 507 includes, or is, a single battery 508. It will be appreciated that a battery can accumulate electric power appropriate for powering one or more sensor and/or actuators, and encompasses various electric power accumulation means, including electrochemical cells, ultracapacitors, and the like. In the example of FIG. 20A, the battery 508 is mounted external to the housing 490, e.g. mounted in or to the derailleur. The battery 508 powers the derailleur 480 as well as the shift actuator 504, which is housed by the housing 490. Electric power may hence be transmitted from the battery 508 to shift actuator 504, via the electric connection interface 921, 922. In the example of FIG. 20B, the battery 508 is housed by the housing 490. The battery 508 powers the shift actuator 504, which is also housed by the housing 490, as well as the derailleur 480, which is external to the housing 490. Electric power may hence be transmitted from the battery 508 to the derailleur 480, via the electric connection interface 921, 922. Hence, the battery 508 may either be arranged internally to the housing 490 or externally to the housing 490, e.g. mounted to a body of the derailleur 480. In the example of FIG. 20C, the derailleur 480 and the shift actuator 504 are powered by respective power supplies, here by respective batteries 508, one of which being arranged internal to the housing 490 and one of which being arranged external to the housing 490, e.g. mounted to a body of the derailleur 480.

[0218] FIG. 21 shows a schematic example of a transmission system 1, similar to the example of FIG. 20B. In this particular example, the transmission system 1 comprises a generator 509. The generator 509 is, here, housed by the housing 490. The generator 509 is configured for converting rotary power of a transmission component of the transmission 907 to electric power. Electric power generated by the generator 509 may be used for powering one or more sensor and/or actuators of

the transmission system **1**, such as in this example the shift actuator **504**, as well as the derailleur **480**, and also an optional control unit **500**. The generator **509** in this example charges the single battery **508** held by the housing **490**. The derailleur **480** and the shift actuator **504** are in this example not directly connected to the generator **509**, but to the battery **508** instead. It will be appreciated that the battery may be omitted in view of the generator. Although in FIG. **21** the generator **509** and battery **508** are both housed in the housing **490**, it will be appreciated that the generator may be housed in housing **490** while the battery is external to the housing, e.g. mounted to the derailleur.

[0219] FIG. **22** shows a schematic example of a transmission system **1**, similar to the example of FIG. **20A**. In this particular example, the generator **509** is mounted external to the housing **490**. The generator **509** can be mounted to the derailleur. The generator **509** is configured for converting rotary power of component of the derailleur, or the wheel, to electric power. Electric power generated by the generator **509** may be used for powering one or more sensor and/or actuators of the transmission system **1**, such as in this example the shift actuator **504**, as well as the derailleur **480**, and also an optional control unit **500**. The generator **509** in this example charges the single battery **508** held by the housing **490**. The derailleur **480** and the shift actuator **504** are in this example not directly connected to the generator **509**, but to the battery **508** instead. It will be appreciated that the battery may be omitted in view of the generator. Although in FIG. **22** the generator **509** and battery **508** are both external to the housing **490**, it will be appreciated that the battery **508** may be housed in housing **490** while the generator **509** is external to the housing, e.g. mounted to the derailleur.

[0220] FIGS. **21** and **22** show schematic examples of a transmission system **1**, wherein the housing **490** furthermore holds an optional control unit **500**. The control unit **500** is here connected to one or more optional external sensors **506**, such as a torque sensor, speed sensor, cadence sensor and/or heart-rate monitor. The control unit **500** is here also connected to an optional external user interface **505**, such as a manual shifter device, e.g. at a handlebar of the bicycle. The control unit **500** may receive a shift command from a user through the user interface **505**, and in turn control the shift actuator **504** and/or the derailleur **480** accordingly. A command signal may hence be transmitted from the user interface **505** to the control unit **500**, e.g. over the electric connection interface **921**, **922**. Also, a sensor signal may be transmitted from the one or more external sensors **506** to the control unit **500**, e.g. over the electric connection interface **921**, **922**. From the control unit **500**, a control signal may be transmitted to the shift actuator **504**. A control signal may also be transmitted from the control unit **500** to the derailleur **480**, e.g. over the electric connection interface **921**, **922**.

[0221] FIGS. **1A** and **1B** show exemplary transmission systems **1**, wherein a continuously variable transmission (CVT) **403** is connected in series with a first transmission **100**. Here, an output of the CVT **403** is connected to an input of the first transmission **100**. However, it is also possible that an output of the first transmission **100** is connected to an input of the CVT **403**. The first transmission **100** is operable according to a first transmission ratio and a second transmission ratio. In this example, the first transmission **100** includes a planetary gear set **50**. In this example, the planetary gear set is configured to selectively provide in a first mode a 1:1 transmission ratio or in a second mode a speed-up or speed-down transmission ratio.

[0222] In FIGS. **1A** and **1B**, the first transmission **100** is connected to a second transmission **200** via an endless drive member **55**, e.g. a chain, belt, or cardan drive. The second transmission can be a hub transmission of the bicycle. The second transmission **200** is operable according to a third transmission ratio and a fourth transmission ratio. The second transmission **200** can e.g. include a second planetary gear set. The second planetary gear set can be configured to selectively provide in a third mode a 1:1 transmission ratio or in a fourth mode a speed-up or speed-down transmission ratio.

[0223] In the exemplary arrangement of FIGS. **1A**, **1B**, the CVT **403** and the first transmission **100** may be configured to be used as a crank transmission, e.g. between a crank and a front chain wheel

of a bicycle. The CVT **403** and the first transmission **100** can e.g. be provided in a common housing, such as a crank housing of a bicycle. The second transmission may be configured to be used as a hub transmission, e.g. between a rear sprocket and wheel hub of the bicycle. The endless drive member **55** may for example connect the front chain wheel and the rear sprocket, for transmitting torque from the front chain wheel to the rear sprocket.

[0224] FIG. **2** shows an example of a transmission system **1**, such as for a two wheeled bicycle. The transmission system **1** includes an input I and an output O. The input I can for example be connected to a crank of the bicycle. The output O can for example be connected to a hub of a driven wheel of the bicycle. Between the input I and the output O, the system includes the first transmission **100**, the CVT **403** and the second transmission **200**. The first transmission **100**, CVT **403** and second transmission **200** are connected to one another in series. A first input **101** of the first transmission **100** is connected to the system input I. A second output **202** of the second transmission **200** is connected to the system output O. A first output **102** of the first transmission **100** is connected to an input **404** of the CVT **403**. An output **405** of the CVT is connected to the second input **201** of the second transmission **200**. It will be appreciated that the CVT output **405** and the second input **201** may be connected to each other via an intermediate member, such as a chain, belt or intermediate shaft as shown in FIGS. **1A**, **1B**. It will be appreciated that also in this example, the CVT may be positioned upstream of the first transmission **100** as shown in FIGS. **1A**, **1B**.

[0225] The first transmission **100** is operable according to a first transmission ratio and a second transmission ratio. Similarly, the second transmission **200** is operable according to a third transmission ratio and a fourth transmission ratio. The first and second transmissions **100**, **200** may include respective gearing, e.g. one or more gears, for providing a reduction or increase transmission ratio between the first input **101** and first output **102**, and between the second input **201** and second output **202**, respectively. The serial arrangement of the first and second transmissions **100**, **200** can thus provide for four distinct system transmission ratios between the system input I and the system output O.

[0226] To shift between the first transmission ratio and the second transmission ratio, the first transmission **100** can include a first clutch, in this example a load-shifting clutch, **C1**. Similarly, the second transmission **100** can include a second clutch, in this example a load-shifting clutch, **C2**, for selectively shifting between the third transmission ratio and the fourth transmission ratio of the second transmission **200**. The first load-shifting clutch **C1** and the second load-shifting clutch **C2** are thus serially arranged between the system input I and the system output O.

[0227] The first transmission **100** has two parallel transmission paths between the first input **101** and first output **102**, namely a first transmission path **100A** and a second transmission path **100B**. At least one of the first and second transmission paths **100A**, **100B** includes the first load-shifting clutch **C1**. Also, at least one of the parallel transmission paths **100A**, **100B** includes a transmission gearing. In this example, the first transmission path **100A** includes a first gearing **R1** arranged for providing the first transmission ratio, and the second transmission path **100B** includes a second gearing **R2** for providing the second transmission ratio.

[0228] Similarly, the second transmission **200** has two parallel transmission paths between the second input **201** and the second output **202**, namely a third transmission path **200A** and a fourth transmission path **200B**. at least one of the third and fourth transmission paths **200A**, **200B** includes the second load-shifting clutch **C2**. Also, at least one of the parallel transmission paths **200A**, **200B** of the second transmission **200** includes a transmission gearing. In this example, the third transmission path **200A** includes a third gearing **R3** arranged for providing the third transmission ratio, and the fourth transmission path **200B** includes a gearing fourth **R4** for providing the fourth transmission ratio.

[0229] The load-shifting clutches **C1** and **C2**, can be used to select an appropriate transmission path between the system input I and system output O. More particular, the first load-shifting clutch **C1**

can be used to selectively switch between the first **100A** and second **100B** parallel transmission paths of the first transmission **100**, and the second load-shifting clutch **C2** can be used to selectively switch between the third **200A** and fourth **200B** parallel transmission paths of the second transmission **200**.

[0230] The load-shifting clutches include at least two states, e.g. a coupled state and a decoupled state, wherein the coupled state couples the clutch input with the clutch output to transmit torque through the clutch, and the decoupled state decouples the clutch input from the clutch output to transmit no torque through the clutch. In the decoupled state, the load shifting clutches **C1**, **C2** enable torque to be transmitted through different, parallel, transmission path.

[0231] In the coupled state of the first load-shifting clutch **C1**, torque can be transmitted through the second transmission path **100B** from the system input **I** to the first output **102**. In the decoupled state, torque can be transmitted through the first transmission path **100A** from the system input **I** to the first output **102**. Similarly, in the coupled state of the second load-shifting clutch **C2**, torque can be transmitted through the fourth transmission path **200B** from the second input **201** to the system output **O**. In the decoupled state, torque can be transmitted through the third transmission path **200A** from the first input **201** to the system output **O**.

[0232] In this example, the load-shifting clutches **C1**, **C2** are provided in, respectively, the second transmission path **100B** and the fourth transmission path **200B**, but it will be appreciated that the first load-shifting clutches **C1**, **C2** can also be provided in, respectively, the first transmission path **100A** and the third transmission path **200A**. The first load-shifting clutch **C1** is here provided between the first input **101** and the second gearing **R2**, but the first load-shifting clutch **C1** can also be provided between the second gearing **R2** and the first output **102**. Similarly, the second load-shifting clutch **C2** is here provided between the second input **201** and the fourth gearing **R4**, but the second load-shifting clutch **C2** can also be provided between the fourth gearing **R4** and the second output **202**.

[0233] Here, the first transmission path **100A** includes a first freewheel clutch **V1**. The first freewheel clutch **V1** can be overrun, e.g. when torque is transmitted through the second transmission path **100B**, e.g. when the first output **102** rotates faster than the first input **101**. Here, the third transmission path **200A** includes a third freewheel clutch **V2**. The third freewheel clutch **V2** can be overrun, e.g. when torque is transmitted through the fourth transmission path **200B**, e.g. when the second output **202** rotates faster than the second input **201**. The freewheel clutches **V1**, **V2** are preferably low friction when overrun to reduce losses.

[0234] Here the freewheel clutches **V1**, **V2** are connected to an input of respective first and third gearing **R1**, **R3**, but it will be appreciated that the freewheel clutches **V1**, **V2** can also be connected to an output of the respective first and third gearing **R1**, **R3**.

[0235] The load-shifting clutches **C1**, **C2** are, at least in this example, particularly arranged to be coupled and decoupled under load, i.e. while torque is transferred through the load-shifting clutch. The load-shifting clutches **C1**, **C2** are for instance form-closed clutches. It will be appreciated that any of the load-shifting clutches may also be force-closed clutches, arranged to transfer torque in at least one rotational direction.

[0236] It will be appreciated that any of the clutches **C1**, **C2** as described herein, e.g. in conjunction with any of the figures, may particularly be a load-shifting clutch as described in WO2018/199757A2, WO2020/085911A2, or WO2021/080431A1, incorporated herein by reference in their entirety.

[0237] Here, the first transmission **100** and the second transmission **200** each comprise a further clutch. In this example, the first transmission **100** comprises a second freewheel clutch **VB1**, and the second transmission **200** comprises a fourth freewheel clutch **VB2**. The second and fourth freewheel clutches **VB1**, **VB2** are, here, connected to the respective inputs of the load-shifting clutches **C1**, **C2**, but it will be appreciated that the freewheel clutches **VB1** and **VB2** can also be connected to the respective outputs of the load-shifting clutches **C1**, **C2**. Alternatively, the

freewheel clutches VB1, VB2 can be omitted, i.e. replaced by solid connections. It may be preferred to connect the second and fourth freewheel clutches VB1, VB2 to the respective inputs of the load-shifting clutches C1, C2, so that the outputs of the load shifting clutches C1, C2 can keep rotating even without inputting any torque through their the inputs. This may facilitate coupling and/or decoupling of the load-shifting clutches C1, C2. The clutches VB1 and VB2 can be particularly arranged to allow for a reverse rotation direction of the output O, i.e. opposite a driving rotation direction, relative to the input I. Here, the first clutch is serially connected in the second transmission path **100B** between the second freewheel clutch VB1 and the second gearing R2. The second clutch is, similarly, serially connected in the fourth transmission path **200B** between the fourth freewheel clutch VB2 and the fourth gearing R4. It will be appreciated that other arrangements are also possible. FIG. 4B shows an exemplary (coaxial) layout of the transmission system **1** of FIG. 4A.

[0238] In this example, the output **102** of the first transmission **100** is connected to an input **404** of the CVT **403**. An output **405** of the CVT **403** is connected to the input **201** of the second transmission **200**. The CVT **403** is arranged to provide a transmission ratio between the CVT input **404** and the CVT output **405**. The CVT **403** can for example be operable according to at least a fifth transmission ratio and a sixth transmission ratio, and additionally a seventh transmission ratio etc. between the input **404** and output **405**. The CVT **403** may hence be arranged to be operable according to various transmission ratios within a continuous range of CVT transmission ratios. The CVT **403** may particularly be arranged to be operable according to a predetermined finite set of transmission ratios within the continuous range of CVT transmission ratios. For example, the CVT **403** may be operable according to any selected one or the predetermined finite set of CVT transmission ratios, such as the fifth transmission ratio, the sixth transmission ratio, etc., The transmission system **1** may comprise a CVT actuator arranged for actuating a transmission ratio change of the CVT.

[0239] In this example, see particularly FIG. 2B, the CVT input **404** is connected to the output **102** of the first transmission **100**, and is hereto provided with gear **100A2** for meshing with gear **100A1** to form the first transmission path **100A**, and with gear **100B2** for meshing with gear **100B1** for forming the second transmission path **100B**. The CVT output **405** is connected to the second input **201** of the second transmission **200**, and is hereto provided with gear **200A1** for meshing with gear **200A2** for forming [0240] the third transmission path **200A** and gear **200B1** for meshing with gear **200B2** for forming the fourth transmission path **200B**.

[0241] The CVT **403** is in this example associated with an intermediate shaft **400**, which is offset from the input shaft I and the output shaft O. The intermediate shaft **400** is in this example be a split shaft, comprising a first intermediate shaft **400A** and a second intermediate shaft **400B** that are rotatable relative to each other. The first transmission **100** is formed between the input shaft I and the first intermediate shaft **400A**. The CVT **403** is configured to apply a transmission ratio, e.g. a seventh and eighth transmission ratio, between the first intermediate shaft **400A** and the second intermediate shaft **400B**. The second transmission **200** is formed between the intermediate shaft **400B** and the output shaft O.

[0242] It will be appreciated that the CVT can be embodied in various ways. In this example, the CVT is a ratcheting type CVT, which uses freewheel or one-way drive modules. However, other types of CVTs can be used such as a pulley-based CVT, a toroidal CVT, a hydrostatic CVT, an electrical CVT, a cone CVT, an epicyclic CVT, a friction-disk CVT, a magnetic CVT, etc., The CVT referred to herein may particularly be a CVT as described in co-pending application NL2028686, which is hereby incorporated by reference in its entirety.

[0243] The transmission system **1** as described thus far is operable according to various transmission ratios, wherein the CVT **403** provides for (pre) programmable transmission ratios. For example, table 1 shows an example of system transmission ratios that is obtainable by the transmission system **1** as shown in FIG. 2A. The example of table shows a seven-speed

transmission system **1** with a substantially constant transmission step size of about 1.25.

TABLE-US-00001 C1 C2 RCVT System transmission ratio R1 = 0.90 Decoupled Decoupled 1.00
0.8 (R1*R3*RCVT) R2 = 1.28 Decoupled Decoupled 1.25 1 (R1*R3*RCVT) R3 = 0.89 Coupled
Decoupled 1.10 1.25 (R2*R3*RCVT) R4 = 1.79 Coupled Decoupled 1.37 1.56 (R2*R3*RCVT)
Decoupled Coupled 1.21 1.95 (R1*R4*RCVT) Coupled Coupled 1.07 2.45 (R2*R4*RCVT)
Coupled Coupled 1.33 3.05 (R2*R4*RCVT)

[0244] In the example of table 1, each consecutive shift changes the system transmission ratio with, approximately, 25%. The transmission ratios of the CVT **403**, RCVT, can be preprogrammed. The CVT can accordingly be controlled to switch from one preprogrammed transmission ratio to another.

[0245] Table 2 shows another example of system transmission ratios that are obtainable by a transmission system **1** as shown in FIG. 2A. This example shows a 10-speed transmission system **1**, with a constant transmission step size of about 1.17.

TABLE-US-00002 TABLE 2 C1 C2 RCVT System transmission ratio R1 = 0.90 Decoupled
Decoupled 1.00 0.80 (R1*R3*RCVT) R2 = 1.28 Decoupled Decoupled 1.17 0.94 (R1*R3*RCVT)
R3 = 0.89 Decoupled Decoupled 1.38 1.10 (R1*R3*RCVT) R4 = 1.79 Coupled Decoupled 1.12
1.28 (R2*R3*RCVT) Coupled Decoupled 1.32 1.50 (R2*R3*RCVT) Decoupled Coupled 1.09
1.75 (R1*R4*RCVT) Decoupled Coupled 1.27 2.05 (R1*R4*RCVT) Coupled Coupled 1.05 2.40
(R2*R4*RCVT) Coupled Coupled 1.23 2.81 (R2*R4*RCVT) Coupled Coupled 1.43 3.29
(R2*R4*RCVT)

[0246] In the example of table 2, each consecutive shift changes the system transmission ratio with, approximately, 17%. The transmission ratios of the CVT **403** (RCVT) can be preprogrammed. The CVT can accordingly be controlled to switch from one preprogrammed transmission ratio to another.

[0247] Table 3 shows another example of system transmission ratios that are obtainable by a transmission system **1** as shown in FIG. 2A. This example shows a 16-speed transmission system **1**, with a constant transmission step size of about 1.10.

TABLE-US-00003 TABLE 3 C1 C2 RCVT System transmission ratio R1 = 0.90 Decoupled
Decoupled 1.00 0.80 (R1*R3*RCVT) R2 = 1.28 Decoupled Decoupled 1.10 0.88 (R1*R3*RCVT)
R3 = 0.89 Decoupled Decoupled 1.21 0.97 (R1*R3*RCVT) R4 = 1.79 Decoupled Decoupled 1.32
1.06 (R1*R3*RCVT) Coupled Decoupled 1.03 1.17 (R2*R3*RCVT) Coupled Decoupled 1.13
1.29 (R2*R3*RCVT) Coupled Decoupled 1.25 1.42 (R2*R3*RCVT) Coupled Decoupled 1.37
1.56 (R2*R3*RCVT) Decoupled Coupled 1.06 1.71 (R1*R4*RCVT) Decoupled Coupled 1.16
1.87 (R1*R4*RCVT) Decoupled Coupled 1.29 2.07 (R1*R4*RCVT) Decoupled Coupled 1.42
2.28 (R1*R4*RCVT) Coupled Coupled 1.09 2.51 (R2*R4*RCVT) Coupled Coupled 1.20 2.76
(R2*R4*RCVT) Coupled Coupled 1.33 3.04 (R2*R4*RCVT) Coupled Coupled 1.45 3.34
(R2*R4*RCVT)

[0248] In the example of table 3, each consecutive shift changes the system transmission ratio with, approximately, 10%. The transmission ratios of the CVT **403** (RCVT) can be preprogrammed. The CVT can accordingly be controlled to switch from one preprogrammed transmission ratio to another.

[0249] Table 4 shows another example of system transmission ratios that are obtainable by a transmission system **1** as shown in FIG. 2A. This example shows a 16-speed transmission system **1**, with a constant transmission step size of about 1.12.

TABLE-US-00004 TABLE 4 C1 C2 RCVT System transmission ratio R1 = 1.11 Decoupled
Decoupled 1.00 0.60 (R1*R3*RCVT) R2 = 1.73 Decoupled Decoupled 1.12 0.67 (R1*R3*RCVT)
R3 = 0.55 Decoupled Decoupled 1.25 0.75 (R1*R3*RCVT) R4 = 1.35 Decoupled Decoupled 1.40
0.84 (R1*R3*RCVT) Coupled Decoupled 1.00 0.94 (R2*R3*RCVT) Coupled Decoupled 1.12
1.06 (R2*R3*RCVT) Coupled Decoupled 1.25 1.18 (R2*R3*RCVT) Coupled Decoupled 1.40
1.33 (R2*R3*RCVT) Decoupled Coupled 1.00 1.49 (R1*R4*RCVT) Decoupled Coupled 1.12

1.66 (R1*R4*RCVT) Decoupled Coupled 1.25 1.86 (R1*R4*RCVT) Decoupled Coupled 1.40
 2.09 (R1*R4*RCVT) Coupled Coupled 1.00 2.34 (R2*R4*RCVT) Coupled Coupled 1.12 2.62
 (R2*R4*RCVT) Coupled Coupled 1.25 3.93 (R2*R4*RCVT) Coupled Coupled 1.40 3.28
 (R2*R4*RCVT)

[0250] In the example of table 4, each consecutive shift changes the system transmission ratio with, approximately, 12%. The transmission ratios of the CVT **403** (RCVT) can be preprogrammed. In this example, the CVT **403** only is preprogrammed to be operable according to four different transmission ratios. The CVT can be controlled to switch from one preprogrammed transmission ratio to another.

[0251] In this example of table 4, the first transmission ratio, here $R1=1.11$, is obtained by a meshing 63-teeth primary gear **100A1** and a 57-teeth secondary gear **100A2**; the second transmission ratio, here $R2=1.73$, is obtained by a meshing 76-teeth primary gear **100B1** and a 44-teeth secondary gear **100B2**; the third transmission ratio, here $R3=0.55$, is obtained by a meshing 78-teeth primary gear **200A1** and a 42-teeth secondary gear **200A2**; and the fourth transmission ratio, here $R4=1.35$, is obtained by a meshing 51-teeth primary gear **200B1** and a 69-teeth secondary gear **200B2**. These gear pairs are such that all summed number of teeth for each meshing gear pair are equal. In this case the summed number of teeth for each meshing gear pair is 120 teeth. The gear pairs are particularly such that a sum of radii of each primary-secondary gear pair is the same for all gear pairs. Hence, the primary gears can be arranged to rotate about a common primary axis, here defined by the coaxial input I and output O shaft, and the secondary gears can be arranged to rotate about a common secondary axis, here defined by the intermediate shaft **400**, parallel to the primary axis, while each pair of gears can be permanently meshingly engaged. The clutches C1 and C2 can accordingly be used to select a desired gear pair for transferring torque, and hence a desired transmission ratio, without having to displace gears. This exemplary first, second, third, and fourth transmission ratios and associated gear teething is summarized in table 5. Other examples are given in tables 6-9.

TABLE-US-00005 TABLE 5 (Gear centre distance: e.g. 60 mm) Primary gear Secondary gear Sum
 Ratio R1 63 teeth 57 teeth 120 teeth 1.11 R2 76 teeth 44 teeth 120 teeth 1.73 R3 78 teeth 42 teeth
 120 teeth 0.54 R4 51 teeth 69 teeth 120 teeth 1.35

TABLE-US-00006 TABLE 6 (Gear centre distance: e.g. 58 mm) Primary gear Secondary gear Sum
 Ratio R1 52 teeth 64 teeth 116 teeth 0.81 R2 65 teeth 51 teeth 116 teeth 1.27 R3 71 teeth 45 teeth
 116 teeth 0.63 R4 46 teeth 70 teeth 116 teeth 1.52

TABLE-US-00007 TABLE 7 (Gear centre distance: e.g. 58 mm) Primary gear Secondary gear Sum
 Ratio R1 52 teeth 64 teeth 116 teeth 0.81 R2 62 teeth 54 teeth 116 teeth 1.15 R3 66 teeth 50 teeth
 116 teeth 0.76 R4 46 teeth 70 teeth 116 teeth 1.52

TABLE-US-00008 TABLE 8 (Gear centre distance: e.g. 58 mm) Primary gear Secondary gear Sum
 Ratio R1 52 teeth 64 teeth 116 teeth 0.81 R2 57 teeth 59 teeth 116 teeth 0.97 R3 66 teeth 50 teeth
 116 teeth 0.76 R4 46 teeth 70 teeth 116 teeth 1.52

TABLE-US-00009 TABLE 9 (Gear centre distance: e.g. 58 mm) Primary gear Secondary gear Sum
 Ratio R1 55 teeth 61 teeth 116 teeth 0.90 R2 68 teeth 48 teeth 116 teeth 1.42 R3 71 teeth 45 teeth
 116 teeth 0.63 R4 46 teeth 70 teeth 116 teeth 1.52

[0252] The examples of tables 1-4 show a relation between the system transmission ratios having a, substantially, constant transmission ratio step between consecutive system transmission ratios. It will be appreciated that any relation can also be obtained using the CVT **403**, e.g. progressively increasing and/or decreasing transmission ratio steps between consecutive system transmission ratios. The CVT **403** can be operated accordingly, e.g. using a control unit.

[0253] Table 10 shows another example of system transmission ratios obtainable by a transmission system as shown in FIGS. 1A, 1B, 2A. This example includes transmission ratios R1, R2, R3, R4, corresponding to the example of table 10. The CVT **403** is preprogrammed to operate according to various transmission ratios RCVT, within a continuous range of transmission ratios.

TABLE-US-00010 TABLE 10 C1 C2 RCVT System transmission ratio R1 = 0.71 Decoupled Decoupled 1.00 0.71 (R1*R3*RCVT) R2 = 1.00 Decoupled Decoupled 1.07 0.76 (R1*R3*RCVT) R3 = 1.00 Decoupled Decoupled 1.15 0.82 (R1*R3*RCVT) R4 = 2.00 Decoupled Decoupled 1.23 0.87 (R1*R3*RCVT) Decoupled Decoupled 1.32 0.94 (R1*R3*RCVT) Coupled Decoupled 1.00 1.00 (R2*R3*RCVT) Coupled Decoupled 1.07 1.07 (R2*R3*RCVT) Coupled Decoupled 1.15 1.15 (R2*R3*RCVT) Coupled Decoupled 1.23 1.23 (R2*R3*RCVT) Coupled Decoupled 1.32 1.32 (R2*R3*RCVT) Decoupled Coupled 1.00 1.42 (R1*R4*RCVT) Decoupled Coupled 1.07 1.52 (R1*R4*RCVT) Decoupled Coupled 1.15 1.63 (R1*R4*RCVT) Decoupled Coupled 1.23 1.75 (R1*R4*RCVT) Decoupled Coupled 1.32 1.88 (R1*R4*RCVT) Coupled Coupled 1.00 2.00 (R2*R4*RCVT) Coupled Coupled 1.07 2.14 (R2*R4*RCVT) Coupled Coupled 1.15 2.30 (R2*R4*RCVT) Coupled Coupled 1.23 2.46 (R2*R4*RCVT) Coupled Coupled 1.32 2.64 (R2*R4*RCVT) Coupled Coupled 1.42 2.83 (R2*R4*RCVT)

[0254] The CVT **403** provides intermediate transmission ratio steps, between the system transmission ratios obtainable with only the first and second transmissions **100**, **200**. Hence, the first and second transmission **100**, **200** combined can provide a spread of system transmission ratios, while the CVT **403** can be used to provide appropriate, intermediate, steps between consecutive ratios. The CVT **403** can also be used to extend the range of system transmission ratios provided by the first and second transmissions **100**, **200**.

[0255] Here, the consecutive system transmission steps are constant, here of 7%, providing a linear set of, here twenty-one, system transmission ratios. It will be appreciated that nonlinear sets may also be obtained by programming the CVT transmission ratios RCVT accordingly. For example, a progressively increasing or decreasing transmission steps may be obtained. The steps may even be changed on-the-fly, i.e. during operation of the transmission, e.g. by properly selecting or re-programming of the CVT transmission ratios RCVT.

[0256] In this example, the transmission ratios are chosen such that the number of different CVT transmission ratios RCVT is less than the number of system transmission ratios. The number of different CVT transmission ratios RCVT is particularly less than half the number of system transmission ratios, more particularly about 25% of the number of system transmission ratios. The CVT **403** is, here, operated according to five different transmission ratios, particularly 1.00, 1.07, 1.15, 1.23 and 1.32, wherein an additional CVT transmission ratio of 1.42 is provided for extending the range provided by the first and second transmissions **100**, **200**.

[0257] Tables 11-13 provide other examples of sets of system transmission ratios obtainable by a transmission system as shown in FIGS. **1A**, **1B**, **2A**, wherein the transmission ratios R1, R2, R3, R4, again correspond to the example of table 10. Compared to table 10, instead of a 21-speed transmission system, tables 11-13 respectively show a 17-speed, 13-speed and 9-speed transmission system. The overall range of system transmission ratios are substantially the same between the examples, here about 400%, but compared to table 10, the CVTs **403** provide fewer intermediate steps.

TABLE-US-00011 TABLE 17 C1 C2 RCVT System transmission ratio R1 = 0.71 Decoupled Decoupled 1.00 0.71 (R1*R3*RCVT) R2 = 1.00 Decoupled Decoupled 1.09 0.77 (R1*R3*RCVT) R3 = 1.00 Decoupled Decoupled 1.19 0.84 (R1*R3*RCVT) R4 = 2.00 Decoupled Decoupled 1.30 0.92 (R1*R3*RCVT) Coupled Decoupled 1.00 1.00 (R2*R3*RCVT) Coupled Decoupled 1.09 1.09 (R2*R3*RCVT) Coupled Decoupled 1.19 1.19 (R2*R3*RCVT) Coupled Decoupled 1.30 1.30 (R2*R3*RCVT) Decoupled Coupled 1.00 1.41 (R1*R4*RCVT) Decoupled Coupled 1.09 1.54 (R1*R4*RCVT) Decoupled Coupled 1.19 1.68 (R1*R4*RCVT) Decoupled Coupled 1.30 1.83 (R1*R4*RCVT) Coupled Coupled 1.00 2.00 (R2*R4*RCVT) Coupled Coupled 1.09 2.18 (R2*R4*RCVT) Coupled Coupled 1.19 2.38 (R2*R4*RCVT) Coupled Coupled 1.30 2.59 (R2*R4*RCVT) Coupled Coupled 1.41 2.82 (R2*R4*RCVT)

TABLE-US-00012 TABLE 18 C1 C2 RCVT System transmission ratio R1 = 0.71 Decoupled Decoupled 1.00 0.71 (R1*R3*RCVT) R2 = 1.00 Decoupled Decoupled 1.12 0.80 (R1*R3*RCVT)

R3 = 1.00 Coupled Decoupled 1.25 0.89 (R1*R3*RCVT) R4 = 2.00 Coupled Decoupled 1.00 1.00 (R2*R3*RCVT) Coupled Decoupled 1.12 1.12 (R2*R3*RCVT) Coupled Decoupled 1.25 1.25 (R2*R3*RCVT) Decoupled Coupled 1.00 1.42 (R1*R4*RCVT) Decoupled Coupled 1.12 1.59 (R1*R4*RCVT) Decoupled Coupled 1.25 1.78 (R1*R4*RCVT) Coupled Coupled 1.00 2.00 (R2*R4*RCVT) Coupled Coupled 1.12 2.24 (R2*R4*RCVT) Coupled Coupled 1.25 2.51 (R2*R4*RCVT) Coupled Coupled 1.40 2.81 (R2*R4*RCVT)

TABLE-US-00013 TABLE 19 C1 C2 RCVT System transmission ratio R1 = 0.71 Decoupled Decoupled 1.00 0.71 (R1*R3*RCVT) R2 = 1.00 Decoupled Decoupled 1.19 0.84 (R1*R3*RCVT) R3 = 1.00 Coupled Decoupled 1.00 1.00 (R2*R3*RCVT) R4 = 2.00 Coupled Decoupled 1.19 1.19 (R2*R3*RCVT) Decoupled Coupled 1.00 1.42 (R1*R4*RCVT) Decoupled Coupled 1.19 1.68 (R1*R4*RCVT) Coupled Coupled 1.00 2.00 (R2*R4*RCVT) Coupled Coupled 1.19 2.37 (R2*R4*RCVT) Coupled Coupled 1.40 2.81 (R2*R4*RCVT)

[0258] FIG. 2B shows an exemplary schematic layout of the transmission system **1** as shown in FIG. 2A. In the example of FIG. 2B, the input shaft and the output shaft are coaxially arranged with respect to one another, but it will be appreciated that an offset arrangement, in which the input shaft and the output shaft are offset with respect to one another, can also be provided. It will further be appreciated that the CVT **403** and/or the input I and/or the output O can be coaxially arranged. Here the CVT **403**, being associated with an intermediate axis **406**, is provided offset from the input and output shafts, to obtain a particular compact setup. The first transmission output **102** and the CVT input **404** are connected to one another. The CVT output **405** and the second transmission input **201** are also connected to one another. A continuously variable transmission ratio may be provided between the CVT input **404** and CVT output **405**. The transmission system of FIG. 2B can e.g. be a crank transmission system. The crank transmission system can be included in a housing at the crank of a bicycle.

[0259] FIG. 3A shows another example of a transmission system **1** comprising a continuously variable transmission (CVT) **403** being connected in series with the first transmission **100** and the second transmission **200**, similar to the example of FIG. 11A. In this example however, alternative to the example of FIG. 11A, the CVT **403** is arranged at an input side of the first transmission **100**, such that the first transmission **100** is arranged between the CVT **403** and the second transmission **200**. The first transmission **100** and the second transmission **200** as shown in FIG. 12A are similar to those shown in e.g. FIG. 11A. Also the CVT **403** may be similar as the CVT **403** shown in FIG. 4A, e.g. a ratcheting-type CVT.

[0260] FIG. 3B shows an exemplary schematic layout of the transmission system **1** as shown in FIG. 3A. In this example, alternative to the example shown in FIGS. 2A and 2B, the CVT input **404** is associated with, e.g. connected to, the system input I of the transmission system **1**, in this example a rotatable input shaft. The input shaft I may for example be attached to a crank of a bicycle. Hence, the input shaft I may be a crank spindle. The transmission shown in FIG. 3B may, for example, be a crank transmission system. The CVT output **405** is connected to the first input **101** of the first transmission **100**. The CVT **403** is arranged to provide a continuously variable transmission ratio, e.g. a set of preprogrammed CVT transmission ratios, between the CVT input **404** and the CVT output **405**. In this example, CVT input **404** is fixed to the input shaft, while the CVT output **405** is rotatable relative to the input shaft. The CVT output **405** is connected to the first input **101** of the first transmission **100**, and is hereto provided with gear **100A1** for meshing gear **100A2** for forming the first transmission path **100A**, and gear **100B1** for meshing with gear **100B2** for forming the second transmission path **100B**.

[0261] The CVT **403** is in this example coaxially arranged with the input shaft I. The CVT is configured to apply a transmission ratio, e.g. a seventh and eighth transmission ratio between the CVT input **404**, here being rotationally fixed to the input shaft I, and the CVT output **405**. The first transmission **100** is formed in this example between the CVT output **405** and an intermediate shaft **400**. The second transmission **200** is formed between the intermediate shaft **400** and the output

shaft **O**. The intermediate shaft **400** is in this example rotatable relative to a stationary mounting shaft **401**. The stationary mounting shaft **401** can for example be mounted to a housing of the transmission system.

[0262] In the example of FIG. 3B, the clutches **C1** and **C2** may for example be clutches as described in WO2018/199757A2, WO2020/085911A2, or WO2021/080431A1. Hence, the clutch **C1** has a first rotatable unit **C1A** connected to the gear **100B1** at the clutch input, and a second rotatable unit **C1B** connected to the gear **100A2** at the clutch output. The first rotatable unit **C1A** includes at least one first abutment surface. The second rotatable unit **C1B** includes at least one second abutment surface arranged for selectively engaging the first abutment surface, the first and second abutment surfaces being adapted to each other so as to allow disengaging under load, preferably in two directions. The clutch **C1** further has a third rotatable unit **C1C**. The third rotatable unit **C1C** includes at least one retaining member, the third unit **C1C** being arranged for selectively being in a first mode or a second mode relative to the second unit **C1B**, wherein the at least one retaining member in the first mode locks the at least one second abutment surface for rotationally coupling the second unit **C1B** to the first unit **C1A**, e.g. in two directions, and in the second mode releases the at least one second abutment surface for decoupling the second unit **C1B** from the first unit **C1A**. In this example, the clutch **C1** comprises a third rotatable unit **C1C** which includes at least one actuation member arranged for moving the third rotatable unit **C1C** from a first position to a second position or from a second position to a first position relative to the second rotatable unit **C1B**. Here, the clutch **C1** further comprises a fourth unit **C1D** including a selector, the selector being arranged for selectively being in a gripping or non-gripping mode; the selector in the gripping mode being arranged for gripping the at least one actuation member for rotating the third rotatable unit **C1C** from the first position to the second position or from the second position to the first position relative to the second rotatable unit; the selector in the non-gripping mode being arranged for not engaging the at least one actuation member. Similarly, the clutch **C2** includes a first rotatable unit **C2A** connected to the input, a second rotatable unit **C2B** connected to the output, and a third rotatable unit **C2C** arranged for co-rotating with the second rotatable unit. The third rotatable unit **C2C** is arranged for selectively being in a first rotational position or a second rotational position relative to the second rotatable unit. The clutch **C2** is arranged for selectively in the first rotational position rotationally coupling the second rotatable unit **C2B** to the first rotatable unit **C2A**, and in the second rotational position decoupling the second rotatable unit **C2B** from the first rotatable unit **C2A**. The clutch **C2** is arranged for temporarily changing rotation speed of the third rotatable unit **C2C** relative to the second rotatable unit **C2A** for rotating from the first position to the second position, or from the second position to the first position.

[0263] The example of FIG. 3A, 3B also includes an optional torque sensor **465**. The torque sensor **465** is arranged to measure a torque transmission through the transmission system **1**, specifically between the first transmission **100** and the second transmission **200**, which is in this example transmitted by the intermediate shaft **400**.

[0264] FIGS. 4A, 4B show respective examples of a transmission system. The CVT **403** is in these example arranged between the system input **I** and the first transmission **100**, as is for example schematically shown in FIG. 3A. The CVT **403** is accommodated within the housing **490**. The CVT input **404** is mounted to the input shaft **I** and rotates therewith. The CVT **403** applies a CVT transmission ratio between the CVT input **404** and the CVT output **405**. The CVT transmission ratio is selectable from a continuous range of transmission ratios. The CVT **403** can for example be arranged to be operable according to a selective one of a finite set of preselected CVT transmission ratios within the continuous range of transmission ratios. A CVT actuator may be provided for actuating a transmission ratio change of the CVT. The CVT output **405** is rotatable about the input shaft **I**, relative to the input shaft **I**. The first input **101** of the first transmission **100** is associated with, or mounted to, the CVT output **405**. Gears **100A1** and **100B1** of the first transmission **100** are in this example mounted to, or integrally formed with, the CVT output **405**, for meshing with

respectively gear **100A2** and gear **100B2**, which are associated and rotatable about the mounting shaft **401**, to form the first and second transmission ratios of the first transmission **100**. Either the first transmission ratio or the second transmission ratio is selected using clutch C1. The second transmission **200** is formed between the mounting shaft **401** and the output shaft O. The third and fourth transmission ratios obtainable with the second transmission **200** are selectable using clutch C2.

[0265] FIGS. 5A, 5B show respective examples of a transmission system similar to the examples shown in FIGS. 4A, 4B further comprising an electric motor **450**. The electric motor **450** is here included in the housing **490**. The electric motor **450** is in this example coupled, here via reduction gearing **451**, **452**, **453**, to the CVT input **404**. The CVT input **404** is in turn coupled to the input shaft I. Hence, the electric motor **450** drives the input shaft I in rotation, via the CVT input **404**. The input shaft I can additionally be driven by an additional power source, such as by muscle power of a user transmitted to the input shaft I via a crank.

[0266] In the example of FIGS. 4A, 4B and 5A, 5B, the clutches C1 and C2 may for example be clutches as described in WO2018/199757A2, WO2020/085911A2, or WO2021/080431A1, as also set out in view of FIG. 2B.

[0267] FIGS. 6A, 6B show an example of a transmission system **1**, similar to the example shown in FIG. 4B. The CVT input **404** is rotatable about a CVT input axis **411**. The CVT input is in this example mounted to the input shaft I, and is hence co-rotates with the input shaft I about the CVT input axis **411**. The CVT output **405** is rotatable relative to the CVT input **405** about a CVT output axis **412**.

[0268] In this example, the CVT output **405** is laterally movable relative to the CVT input **404**, in a direction transverse to the CVT input axis **411**. Various CVT transmission ratios can be obtained by laterally moving the CVT output **405** relative to the CVT input **404**. The CVT input **404** and the CVT output **405** are in this example coupled by coupling elements, arranged for transferring torque from the CVT input **404** to the CVT output **405** at different radii. For example, the coupling elements may be provided at a constant first radius from the CVT input axis and at a variable second radius from the CVT output axis, wherein the second radius is variable by moving the CVT output **405** laterally relative to the CVT input **404**, so as to offset the CVT output axis **412** from the CVT input axis **411**. An example of such CVT is described in detail in co-pending patent application NL2028686.

[0269] FIG. 6A shows the transmission system **1** in a first state, in which the CVT input axis **411** and the CVT output axis **412** coincide. In the first state, the CVT **403** operates according to a 1:1 transmission ratio. FIG. 6B shows the transmission system **1** in a second, different state, in which the CVT output axis **412** is offset from the CVT input axis **411**. The transmission system is transitioned from the first state to the second state by moving the CVT output **405** relative to the CVT input **404** in a direction transverse to the CVT input axis **411**.

[0270] In this example, the CVT output **405** is pivotably drivable about a parallel axis; parallel to the CVT input axis **411**. In this particular example, the CVT output **405** is pivotably coupled to the stationary mounting shaft **401**, by a pivot arm **415**. The pivot arm extends here between the mounting shaft **401** and the CVT output **405**. The CVT output **405** is pivotably drivable at a constant radius from the mounting shaft **401**; the constant radius being defined by the pivot arm **415**. The primary gears **100A1** and **100B1** of the first transmission **100** are in this example mounted to the CVT output **405** and are hence also pivotable along with the CVT output **405** about the mounting shaft **401**. Because the secondary gears **100A2** and **100B2** of the first transmission are rotatably associated with the mounting shaft **401**, the primary-secondary gear pairs **100A1-100A2** and **100B1-100B2** can remain meshingly engaged while the CVT output **405** is being pivoted about the mounting shaft **401**.

[0271] A CVT shift actuator may be arranged for pivoting the pivot arm about the mounting shaft **401**, in turn offsetting the CVT output **405** relative to the CVT input **404**. The pivot arm **415** may

for example be fixed to the mounting shaft **401**, wherein the CVT shift actuator is arranged for rotating the mount shaft **401** about its longitudinal axis.

[0272] FIGS. 7A, 7B show an example of a gearless transmission system **1**. The transmission system **1** in this example is similar to the example shown in FIGS. 6A, 6B, but instead of gear drives for the first and second transmissions **100**, **200**, the first and second transmissions includes belt drives for transmitting torque. In particular, the first transmission **100** comprises a first endless drive member **110A** arranged in the first transmission path **100A**, and a second endless drive member **110B** arranged in the second transmission path **100B**. The first and second endless drive members **110A**, **110B**, e.g. a first and second belt or chain, connect, respectively, a first and second primary wheel **100A1**, **100B1**, e.g. a primary chain wheel, with a first and second secondary wheel **100A2**, **100B2**, e.g. a secondary chain wheel.

[0273] In this example, the second transmission **200**, similarly, comprises a third endless drive member **210A** arranged in the third transmission path **200A**, and a fourth endless drive member **210B** arranged in the fourth transmission path **200B**. The third and fourth endless drive members **210A**, **210B**, e.g. a third and fourth belt or chain, connect, respectively, a third and fourth primary wheel **200A1**, **200B1**, e.g. a primary chain wheel, with a third and fourth secondary wheel **200A2**, **200B2**, e.g. a secondary chain wheel. It will be appreciated that the any of the gear drives of the transmission systems described herein may also be configured as a belt drive.

[0274] The exemplary gearless transmission system **1**, shown in FIGS. 7A, 7B, does not include meshing gears, and may accordingly be free of oil lubrication. No oil bath is present for lubricating rotational members of the transmission. Standard bearings, e.g. roller bearings, are used instead. There are in this example accordingly no need for oil seals to seal the housing **290**. Instead of lubrication oil, a minimal amount of lubrication grease may be applied. The endless drive members and their associated wheels may even be completely free of lubrication.

[0275] FIGS. 8A-8B show exemplary distributed transmission systems **1**, wherein the CVT **403** and the first transmission **100** are held by a first housing **490**, here a crank housing **490** as the CVT **403** and the first transmission **100** together here operate as a crank transmission, and the second transmission **200** is held by a second housing, here a hub housing as the second transmission here operate as a hub transmission. The crank transmission system, here comprising the CVT **403** and the first transmission **100**, is connected to the hub transmission system, here comprising the second transmission **200**, via an endless drive member **55**, here a chain. The CVT **403** is connected at an input side of the first transmission **100**, wherein, here, the first transmission **100** includes a planetary gear set **50**. The planetary gear set is arranged in one of the transmission paths of the first transmission **100**, here in the second transmission path **100B**. The other transmission path, here the first transmission path **100A**, is in this example provides a 1:1 transmission ratio. The first transmission **100** is, here, connected to the second transmission **200** via the endless drive member **55**, e.g. a chain or belt, but it will be appreciated the second transmission **200** and the endless drive member **55** may also be omitted. In the exemplary arrangement of FIGS. 8A, 8B, the CVT **403** and the first transmission **100** may be configured to be used as a crank transmission, e.g. between a crank and a front chain wheel of a bicycle, and the second transmission may be configured to be used as a hub transmission, e.g. between a rear sprocket and wheel hub of the bicycle. The endless drive member **55** may for example connect the front chain wheel and the rear sprocket, for transmitting torque from the front chain wheel to the rear sprocket.

[0276] In the example of FIG. 8A, the planet gear rotation axes are parallel to the input axis **411**, whereas in the example of FIG. 8B the planet gear rotation axes are arranged at an angle, particularly transverse, to the input axis **411**.

[0277] The planet gears **53** in FIG. 8A are embodied as stepped planet gears. This way, a transmission ratio obtainable with the planetary gear can be increased, compared to arrangement with non-stepped planet gears. Hence, each planet gear **53** includes a large planet member **53A** and a small planet member **53B** being fixed to each other, and co-rotatable about a respective rotation

axis. The large planet member **53A** of the stepped planet gear engages the ring gear **54**, whereas the small planet member **53B** engages the sun gear **51**.

[0278] The first transmission **100** is selectively operable according to two different transmission ratios **R1**, **R2**, wherein, here, **R1**=1.00 and **R2**=2.00. **R2** is provided by the planetary gear set. With clutch **C1**, the first transmission **100** can switch between the first transmission path **100A** and the second transmission path **100B**. Clutch **C1** can either be connected at an input of the planetary gear set, or at an output of the planetary gear set.

[0279] Torque is transmitted from the CVT input **404** to the CVT output **405**. The planet carrier **52** is fixed to the CVT output **205** and corotates therewith about the input shaft I. From the planet carrier **52**, torque is transmitted via the stepped planet gears **53** to the sun gear **51**, which is couplable to the first transmission output **102** via clutch **C1**. If clutch **C1** is decoupled, however, torque is not transmitted through the planetary gear set, but through the first transmission path **100A** which bypasses the planetary gear set. Via the first transmission path **100A**, torque is transmitted from CVT output **405** to the first transmission output **102** via freewheel clutch **V1**. The output **102** of the first transmission **100** may be connected to the input **201** of the second transmission **200**, here via an endless drive member **55**, e.g. a chain or belt. Another connections are also envisioned.

[0280] FIGS. **9A** and **9B** show exemplary transmission systems **10**, particularly in accordance with the example of FIGS. **8A** and **8B**. In FIGS. **9A**, **9B** an exemplary CVT **403** is shown. Here, the CVT comprises a first drive element **410** that is rotatable about a first axis. Here the first axis coincides with the shaft I. The CVT further comprises a second drive element **420A**, **420B**, **400** that is rotatable about a second axis parallel to the first axis, wherein the second drive element is movable relative to the first drive element in a direction transverse to the first axis. The CVT comprises first coupling elements **411** for transferring torque between the first drive element **410** and the second drive element **420A**, wherein the first coupling elements are provided at a constant first radius from the first axis and at a variable second radius from the second axis. Such CVT is e.g. described in co-pending application NL2028686 incorporated herein by reference in its entirety. In this example, the CVT comprises a third drive element **430** that is rotatable about a third axis parallel to the second axis. Here, the third axis also coincides with the shaft I. The third drive element **430** and the second drive element **420B** are movable relative to each other in a direction transverse to the third axis; and second coupling elements **421** provided at a constant third radius from the third axis and at the variable second radius from the second axis, for transferring torque between the third drive element **430** and the second drive element **420B**. Such CVT is e.g. described in co-pending application NL2028686 incorporated herein by reference in its entirety. It will be appreciated that such CVT can also be used in the other examples shown herein. It will be appreciated that the CVT of FIG. **7B** can be a CVT with the first drive element **410** and the second drive element **420A**, but without the third drive element **430**.

[0281] FIGS. **10A** and **20B** show examples of a transmission system **10**, which may be offset alternative configurations to the examples shown in FIGS. **9A-9B**. FIGS. **10A** and **10B** differ from each other in the configuration of the CVT unit **403**, in that in FIG. **10A** the second drive member parts **420A**, **420B** are connected at a radius larger than where the coupling elements **411**, **421** are present, while in FIG. **10B** the second drive member parts **420A**, **420B** are connected at a radius smaller than where the coupling elements **411**, **421** are present.

[0282] Similar to the example of FIGS. **9A-9B**, one transmission path of the first transmission **100** provides a 1:1 transmission ratio between the third drive element **430** and the output O, and another transmission path provides a non-unity transmission ratio, here a transmission ratio of 2.00. Instead of a coaxially arranged planetary gear set **50** as shown in FIGS. **9A** and **9B**, the second transmission path **100B** extends via an offset intermediate axis, here defined by a stationary mounting shaft **401**. Compared to the coaxial arrangement shown in FIGS. **9A**, **9B**, the examples of FIGS. **10A** and **10B** are relatively compact in axial direction. On the other hand, the coaxial

arrangement of FIGS. 9A-9B is relatively compact in radial direction, compared to the offset arrangement of FIGS. 10A and 10B.

[0283] FIGS. 11A-11C show an exemplary implementation of a transmission system **1** as described herein as a crank transmission for a bicycle. FIGS. 11A-11C show particularly a crank assembly **10**, wherein the transmission system **1** is provided between a crank **60** of the bicycle and a (front) chain wheel **62**. FIG. 11A shows a top view of the crank assembly **10**. FIG. 11B shows a side view of the crank assembly **10**. FIG. 11C shows a cross-sectional view of the crank assembly **10**.

[0284] In this example, the crank **60** is connected to, or forms, the system input **I** and is connected to the front chain wheel via the CVT **403** and the first transmission **100**, and optionally via the second transmission **200**. The front chain wheel **62** can engage an endless drive member, e.g. a chain or belt. Hence, the transmission system **1** can selectively provide one of various transmission ratios between the crank **60** and the front chain wheel **62**.

[0285] The crank **60** and the front chain wheel **62** are associated with respectively an input shaft and an output shaft, which are (coaxially) rotatable about a common crank axis **407**. The crank **60** and the front chain wheel **62** are connected via an intermediate shaft **400**, which can be rotatable about an intermediate axis **406** parallel offset to the crank axis **407**.

[0286] An electric motor **450**, here a brushless DC motor, is arranged for driving the front chain wheel **62** additionally or alternatively to the crank **62**. The electric motor **450** can be connectable to any one of the input shaft, intermediate shaft, and output shaft. In this example, the electric motor **450** is connected to the input shaft, here via pinion **453**. Here, separate gearing **451**, **452** is provided between the electric motor **450** and the intermediate shaft, comprising gear **451** mounted on the input shaft, and gear **452** mounted on the intermediate shaft. Hence, the transmission system can provide different transmission ratios between the electric motor and the front chain wheel **62** and between the crank **60** and the front chain wheel **62**. From the intermediate shaft torque inputted by the electric motor **450** is transmitted through a transmission path of the transmission system e.g. through a transmission path of the second transmission.

[0287] Here, an output axis **408** of the electric motor **450** extends transverse to the crank axis **407**. The electric motor may be accommodated within a bicycle frame, for example in a seat tube or down tube of a bicycle frame. In this example, the output axis **408** extends radially with respect to the crank axis **407**, in a direction that corresponds to a direction in which the downtube of a bicycle extends. The electric motor **450** in this example particularly has a cylindrically shaped housing for being accommodated in the down tube. The intermediate shaft is angularly spaced from the electric motor **450** to provide a compact setup.

[0288] FIG. 12 shows a transmission system **1**, similar to that as shown in FIG. 2A. FIG. 13 shows a transmission system, similar to that as shown in FIG. 3A. Here, the transmission systems **1**, as shown in FIGS. 12 and 13, includes a control unit **500**. It will be appreciated that each of the examples described herein may include a control unit **500** configured to receive a first shift signal and a second shift signal, and configured to control the first load-shifting clutch and/or the second load-shifting clutch and/or the third load-shifting clutch for selectively coupling or decoupling in response to receiving the first and/or second shift signal. The shift signal may for example be sent, e.g. wirelessly, from a user interface **505**, such as from a manual shifter device, e.g. at a handlebar of the bicycle, and/or one or more sensors **506**, such as a torque sensor, speed sensor, cadence sensor and/or heart-rate monitor.

[0289] The first shift signal can be an upshift signal and the second shift signal can be a downshift signal. The control unit **500** can be configured to selectively control the first and/or second and/or third load-shifting clutch (and optionally the CVT) for selecting the next higher system transmission ratio in response to receiving the upshift signal, and for selecting the next lower system transmission ratio in response to receiving the downshift signal. The controller can also be configured to selectively control the first and/or second and/or third load-shifting clutch for selecting the second next, third next, fourth next, fifth next, sixth next, seventh next, eighth next

higher or lower system transmission ratio in response to receiving a bail-out signal. The bail-out signal may for instance include the upshift signal and downshift signal at the same time, or within a specified time-interval.

[0290] The control unit **500** can thus be connected, e.g. wirelessly, to a first actuator **501** for actuating the first, e.g. load-shifting, clutch **C1** and to a second actuator **502**, e.g. wirelessly, for actuating the second, e.g. load-shifting, clutch **C2**. The control unit **500** can also be connected, e.g. wirelessly, to a third actuator **503** for actuating the CVT **403**. The CVT **403** may be operable according to various transmission ratios. Each of the CVT transmission ratios may preprogrammed, and adapted to the transmission ratios of the first and second transmissions, **100**, **200**. A power supply **507** may supply power, e.g. electric power, to the control unit **500** and the actuators **501**, **502**, **503**, sensors **506**, and/or the user interface **505**. The power supply may for example comprise a battery. The control unit **500** may also be arranged to operate the electric motor **450**. The control unit **500** may be configured to regulate an output power or output torque of the electric motor **450**. The control unit **500** may also be configured to operate a clutch for coupling and decoupling the electric motor **450** from transmission system. The electric motor **450** may be powered from a separate power source. The control unit **500** may for example include a look-up table to synchronize actuation of the one or more actuators in response to a shift signal.

[0291] FIG. **14** shows an example, similar to FIG. **13**, where the CVT **403** and the first transmission **100** are included in a first housing **490**, and wherein the second transmission **200** is included in a separate second housing **491**. Here, the first housing **490** corresponds to a crank housing, and the CVT **403** and the second transmission **200** act as a crank transmission. Here, the second housing **491** corresponds to a hub housing, e.g. partly formed by a hub shell of a driven wheel, and the second transmission acts as a hub transmission. It will be appreciated that, alternatively, the first housing may correspond to a hub housing and the second housing to a crank housing. In this example, the first housing **490** also includes the control unit **500**, the power supply **507**, the electric motor **450**, and the actuators **501**, **503** for actuating the first transmission **100** and the CVT. The second housing **491** includes in this example the actuator **502** for actuating the second transmission **200**. The actuator **502** held by the second housing **491** is connected to the control unit **500** held by the first housing **491**, via a wired or wireless connection. Here, the actuator **502** is powered by the power supply **507** held by the first housing **490**, wherein, in this example, the power is supplied from the power supply **507** to the actuator **502** via the control unit **500**.

[0292] FIG. **24** shows a bicycle **1000**. The bicycle **1000** comprises a frame **1002** with a front fork **1005** and a rear fork **1007**, as well as a front wheel and a rear wheel **1011**, **1013** located in the front and rear fork respectively. The bicycle **1000** further comprises a crank **1017**, and a front chain wheel **1019**. In this example, the CVT **403** and the first transmission **100** are interconnected between the crank **1017** and front chain wheel **1019**. The bicycle **1000** also comprises a rear sprocket **1021** and a rear wheel hub **1022** of the rear wheel **1013**, wherein a chain **1023** threads over the front chain wheel **1019** and rear sprocket **1021**. In this example, the second transmission **200** is interconnected between the rear sprocket **1021** and the rear wheel hub **1022**. A CVT **403** may be arranged between the crank **1017** and the front chain wheel **1019**, particularly between the crank **1017** and the first transmission **100**. The bicycle **1000** also comprises a control unit **500**, here connected to handlebars **1031**. Here, the bicycle **1000** does not comprise a front and rear derailleur.

[0293] Herein, the invention is described with reference to specific examples of embodiments of the invention. It will, however, be evident that various modifications and changes may be made therein, without departing from the essence of the invention. For the purpose of clarity and a concise description features are described herein as part of the same or separate embodiments, however, alternative embodiments having combinations of all or some of the features described in these separate embodiments are also envisaged.

[0294] In the claims, any reference signs placed between parentheses shall not be construed as

limiting the claim. The word 'comprising' does not exclude the presence of other features or steps than those listed in a claim. Furthermore, the words 'a' and 'an' shall not be construed as limited to 'only one', but instead are used to mean 'at least one', and do not exclude a plurality. The mere fact that certain measures are recited in mutually different claims does not indicate that a combination of these measures cannot be used to an advantage.

Claims

1. A wheel hub assembly comprising a hub housing holding a shift actuator, and an electric connection interface for electrically connecting the shift actuator with an externally arranged electric component of an electronic derailleur, the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the externally arranged electric component and the shift actuator.
2. The wheel hub assembly of claim 1, wherein the shift actuator and the electronic derailleur are powered by a common electric power source.
3. The wheel hub assembly of claim 2, wherein the common electric power source includes a generator and/or a battery.
4. The wheel hub assembly of claim 3, wherein the electronic derailleur includes the battery being mounted to a derailleur body, which battery is connected to power the electronic derailleur and the shift actuator held by the hub housing.
5. The wheel hub assembly of claim 3, wherein the hub housing holds the battery, and wherein the electronic derailleur is connected to be powered by the battery held by the hub housing.
6. The wheel hub assembly of claim 3, further comprising an axle, wherein the common electric power source includes the battery, wherein the battery is mounted to or held by the axle for powering the shift actuator held by the hub housing and the electronic derailleur.
7. The wheel hub assembly of claim 3, wherein the electronic derailleur includes the generator.
8. The wheel hub assembly of claim 3, wherein the common electric power source includes the generator and wherein the hub housing holds the generator.
9. The wheel hub assembly of claim 1, wherein the hub housing includes a driver for being connected to a torque transfer member and a hub shell for being connected to a driven wheel, wherein the wheel hub assembly further comprises a hub transmission being selectively operable according to at least two different transmission ratios, the hub transmission being connected between the driver and the hub shell, wherein the shift actuator is configured for operating the hub transmission for switching from one transmission ratio to another transmission ratio.
10. (canceled)
11. The wheel hub assembly of claim 1, wherein the electric connection interface is configured for connection to an external control unit, an external battery, an external user-operable shifter and an external transmission.
12. The wheel hub assembly of claim 1, wherein the electric connection interface comprises; a first connection part having a first electric contact being electrically connected to the shift actuator; and a second connection part, being releasable connectable to the first connection part, and having a second electric contact arranged for being electrically connected to an external electric component, wherein the first and second electric contacts are arranged for being in contact with one another so as to conduct electric signals therebetween when the first and second connection parts are mechanically coupled to each other.
13. (canceled)
14. The wheel hub assembly of claim 12 any of claims 12-13, comprising a third connection part arranged for being mechanically and electrically connected, or releasably or, e.g. releasably, connectable, to the second connection part.
15. (canceled)

- 16.** The wheel hub assembly of claim 14, wherein the third connection part includes a wireless transmitter and/or receiver for providing a wireless communication channel between the third connection part and the external electric component.
- 17.-23.** (canceled)
- 24.** The wheel hub assembly of claim 12, further comprising a driver for being connected to a torque transfer member and a hub shell for being connected to a driven wheel and a bearing arranged between the driver and a wheel axle for allowing the driver to rotate relative to the wheel axle about a wheel axis, wherein the first connection part is arranged, in radial direction, between the bearing and the wheel axle.
- 25.-29.** (canceled)
- 30.** A crank assembly for a bicycle, comprising a crank housing holding a shift actuator, and an electric connection interface for electrically connecting the shift actuator with an externally arranged electric component, the electric connection interface being arranged for transmitting electric power and/or electronic communication signals between the externally arranged electric component and the shift actuator.
- 31.** The crank assembly of claim 30, comprising an input shaft connected to a crank and an output shaft connected to a torque transfer member, and a crank transmission, being selectively operable according to at least two different transmission ratios, the crank transmission being connected between the input shaft and the output shaft, wherein the shift actuator is configured for operating the crank transmission for switching from one transmission ratio to another transmission ratio.
- 32.** A crank assembly of claim 31, wherein the input shaft and the output shaft are rotatable around a common drive axis, and wherein the crank assembly comprises an electric motor connected to the input shaft or the output shaft, wherein the electric motor has a rotatable output member that is rotatable about an electric motor output axis which extends transverse to the common drive axis.
- 33.** A transmission system for a human powered vehicle, such as a bicycle, comprising: an electronically actuatable hub transmission having a first electronic actuator for selecting one of at least two different selectable transmission ratios; an electronically actuatable derailleur having a second electronic actuator for selecting one of a plurality of sprockets; and a common electric power source for providing electric power to both the first electronic actuator and second electronic actuator.
- 34.** The transmission system of claim 33, wherein the common electric power source includes a generator and/or a battery.
- 35.** The transmission system of claim 34, wherein the battery is included by the electronically actuatable derailleur, a hub housing, or an axle.
- 36.-37.** (canceled)
- 38.** The transmission system of claim 34, wherein the generator is included by the electronically actuatable derailleur, or a hub housing.
- 39.-42.** (canceled)
- 43.** The wheel hub assembly of claim 1, wherein the hub housing further holds one or more sensors.
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